Online Resource 3

Per-article table of the 110 coded records

Supplementary material to: *How the Classical Baseline Shapes Machine-Learning Speed Claims in Topology Optimization.* Shaoliang Yang and Jun Wang, Santa Clara University. Corresponding author: Jun Wang, jwang22@scu.edu. Results in Engineering.

Reading guide. Rules R1–R4 assess quality matching, classical efficiency, offline cost and verification of returned designs, respectively. ML/AI, machine learning/artificial intelligence; TO, topology optimization; FE/FEM/FEA, finite-element method/analysis; CNN/DNN/ANN/NN, convolutional/deep/artificial/neural network; GAN, generative adversarial network; VAE, variational autoencoder; CPU/GPU, central/graphics processing unit; DOF, degree of freedom; SIMP, solid isotropic material with penalization; OC, optimality criteria; MMA, method of moving asymptotes; MSE/RMS/MAE/MAPE, mean squared/root mean square/mean absolute/mean absolute percentage error; IoU, intersection over union. Article-specific labels and locators in the evidence refer to the cited article. The supplied CSV retains the exact working-record fields. Stratum Q denotes the quality-only set; C3 is the protocol's offline-cost question, assessed by R3; na means not applicable (a dash in the table). R-prefixed markers are the protocol's recorded reason labels. Verdicts are the adjudicated verdicts before the authors' review; the seven review changes are logged in Online Resource 1, and the paper reports the review count (11 of 28) separately. COI marks a record co-authored by the authors (conflict of interest); ESM_3.csv is the spreadsheet supplied with this Online Resource. Author–year labels follow the bibliographic records used for sampling and are retained here. The manuscript cites confirmed journal versions where available and preserves the coded preprint identifiers in its bibliography. Differences in names, publication years, or DOIs between this table and the reference list do not change which version was coded: the original source identifier and record ID govern the audit evidence.

Reader's note. The evidence texts below are the coders' working records, reproduced as coded and hashed before analysis. Terms such as fail, partial, loose or inefficient describe how an article's reported comparison scores against the audit rules of Online Resource 1; they are not judgements of the article's scientific merit or of its authors' intent. Bracketed notes are the coders' own: [inferred] and [derived] mark statements reasoned or computed from the text, [read from plot] and similar mark values read from a figure or table, [judgment] marks a call the rules leave open, and [recalled, not verified in-session] marks a statement a coder made from memory without checking a source.

All 110 coded records of the analysis sample (positions 1–200 of the seeded order, excluding records listed as paywalled). The complete fields, including the evidence text with locators for every verdict, are in the spreadsheet `ESM_3.csv`, supplied with this document. Section 1 lists every record; Sect. 2 gives the verdict details of the included articles. Verdicts: P pass, p partial, F fail, – not applicable. Stage: agreed (the two blind codings agreed) or adjudicated. Record P01900 carries a conflict-of-interest marker (see the paper's declarations).

Local benchmark-log paths and the record hash manifest identify artifacts in the research repository; they are not files in this PDF/CSV bundle.

1. All coded records

Record	Article	DOI	Decision	Stratum	R1	R2	R3	R4	Stage
P00003	Nguyen et al. (2026)	10.48550/arXiv.2608.22606	include	quality	P	–	–	p	adjudicated
P00006	Joo et al. (2026)	10.48550/arXiv.2607.00438	include	quality	p	–	–	p	adjudicated
P00019	Sun et al. (2026)	10.48550/arXiv.2602.17783	include	quality	p	–	–	p	adjudicated
P00029	Singh et al. (2025)	10.48550/arXiv.2511.04795	include	speed	p	P	F	p	agreed
P00034	Lutheran et al. (2025)	10.48550/arXiv.2509.05800	exclude	none					adjudicated
P00042	Tabarraei (2025)	10.1007/s00366-025-02208-x	exclude	none					agreed
P00051	Radler et al. (2025)	10.48550/arXiv.2502.13174	include	quality	p	–	–	p	adjudicated
P00080	Padhy et al. (2024)	10.1007/s00366-024-02077-w	include	speed	p	p	P	p	adjudicated
P00081	Chadha and Kumar (2024)	10.48550/arXiv.2404.12244	include	quality	p	–	–	p	adjudicated
P00092	Zhong et al. (2023)	10.1007/s00158-023-03686-7	include	quality	p	–	–	p	adjudicated
P00136	Gladstone et al. (2021)	10.48550/arXiv.2107.10661	include	quality	p	F	–	p	adjudicated
P00139	Dupuis and Jacot (2021)	10.48550/arXiv.2106.05710	exclude	none					agreed
P00140	Elingaard et al. (2021)	10.1016/j.cma.2021.114197	include	speed	p	P	p	p	adjudicated
P00141	Behzadi and Ilieş (2021)	10.48550/arXiv.2105.03045	include	quality	p	–	–	p	adjudicated

Record	Article	DOI	Decision	Stratum	R1	R2	R3	R4	Stage
P00143	Bhattacharya et al. (2021)	10.48550/arXiv.2104.04017	exclude	none					adjudicated
P00148	Deng and To (2021)	10.48550/arXiv.2101.03286	include	quality	p	–	–	p	adjudicated
P00149	Zheng et al. (2021)	10.1016/j.cma.2021.113894	include	speed	P	P	p	p	adjudicated
P00150	Halle et al. (2021)	10.3390/app11199041	include	speed	p	P	P	p	adjudicated
P00161	Nakamura and Suzuki (2020)	10.48550/arXiv.2004.05461	include	speed	F	P	F	p	adjudicated
P00165	Abueidda et al. (2020)	10.1016/j.compstruc.2020.106283	include	quality	p	–	–	F	adjudicated
P00170	Jiang et al. (2019)	10.1016/j.advengsoft.2020.102841	exclude	none					agreed
P00176	Zhang et al. (2019)	10.48550/arXiv.1901.07761	include	speed	F	P	F	P	adjudicated
P00178	Rawat and Shen (2019)	10.48550/arXiv.1901.04859	include	speed	F	p	F	F	agreed
P00181	Banga et al. (2018)	10.48550/arXiv.1808.07440	include	speed	p	P	F	F	adjudicated
P00183	Cang et al. (2019)	10.1016/j.cad.2018.12.008	include	speed	F	p	p	P	adjudicated
P00193	Kwon and Kang (2026)	10.48550/arXiv.2609.10643	include	quality	p	F	–	P	adjudicated
P00238	Zhao et al. (2026)	10.48550/arXiv.2603.27195	exclude	none					adjudicated
P00309	Xu et al. (2024)	10.1016/j.matdes.2025.114380	include	quality	p	–	–	p	adjudicated
P00313	Yan et al. (2025)	10.1016/j.cma.2025.117950	exclude	none					adjudicated
P00335	Feng et al. (2024)	10.48550/arXiv.2406.16066	exclude	none					agreed
P00358	Krishnamurthy (2024)	10.1016/j.cad.2024.103707	include	quality	p	–	–	F	adjudicated
P00373	Mallon et al. (2023)	10.1002/nme.70004	include	quality	p	–	–	p	adjudicated
P00375	Giannone and Ahmed (2023)	10.1115/detc2023-116595	include	quality	p	–	–	p	adjudicated
P00519	Geubelle (2025)	10.1016/j.cma.2025.118316	exclude	none					adjudicated
P00523	Hashemi et al. (2021)	10.1016/j.commatsci.2021.110664	exclude	none					adjudicated
P00527	Jeong et al. (2023)	10.1016/j.engstruct.2022.115484	include	quality	P	–	–	p	adjudicated
P00533	Jeong et al. (2023)	10.1016/j.cma.2023.116401	include	none	–	–	–	–	agreed
P00537	Li et al. (2019)	10.1016/j.cad.2019.05.038	include	none	–	–	–	–	agreed
P00549	Kudela et al. (2023)	10.1016/j.ymssp.2023.110636	include	speed	p	p	p	p	adjudicated
P00550	Senhora et al. (2022)	10.1016/j.cma.2022.115116	include	speed	p	P	F	p	adjudicated
P00597	Sun et al. (2022)	10.1016/j.matdes.2022.110885	include	none	–	–	–	–	agreed
P00601	Gavris and Sun (2024)	10.1016/j.eml.2024.102215	include	none	–	–	–	–	agreed
P00605	Liu et al. (2023)	10.1111/mice.13100	include	none	–	–	–	–	agreed
P00610	Adam et al. (2023)	10.1016/j.egyai.2023.100310	include	speed	P	p	P	P	adjudicated
P00624	Rade et al. (2023)	10.1016/j.engappai.2023.107033	include	none	–	–	–	–	agreed
P00647	Viswanath et al. (2024)	10.3389/fmech.2024.1417606	include	speed	p	p	P	F	agreed
P00648	Brown et al. (2022)	10.1016/j.matdes.2022.110672	include	none	–	–	–	–	adjudicated
P00663	Eigel et al. (2022)	10.3390/a15070241	include	speed	P	P	F	p	agreed
P00676	Suresh (2021)	10.48550/arxiv.2109.01861	include	quality	p	–	–	p	adjudicated
P00698	Kong et al. (2024)	10.3390/app142311416	include	speed	p	P	p	p	adjudicated
P00705	Ullah et al. (2024)	10.3390/jmmp8050220	include	speed	F	p	p	F	agreed
P00707	Xing and Tong (2025)	10.1016/j.engstruct.2025.120507	include	none	–	–	–	–	agreed
P00709	Gao et al. (2025)	10.1016/j.ymssp.2025.113670	include	none	–	–	–	–	adjudicated
P00712	Kallioras and Lagaros (2020)	10.1016/j.promfg.2020.02.273	include	none	–	–	–	–	agreed
P00740	Keshavarzzadeh et al. (2021)	10.1016/j.cad.2020.102947	include	speed	P	P	P	P	adjudicated
P00755	Yu et al. (2023)	10.1016/j.compbiomed.2022.106475	include	speed	F	P	p	P	adjudicated
P00786	Luo et al. (2025)	10.1093/jcde/qwaf064	include	speed	P	P	p	p	adjudicated
P00803	Deng et al. (2020)	10.48550/arXiv.2002.01927	exclude	none					agreed
P00808	Zhang et al. (2026)	10.3390/sym18030537	include	speed	p	P	–	p	adjudicated
P00822	Pereira and Driemeier (2024)	10.1016/j.istruc.2024.106924	include	none	–	–	–	–	agreed
P00827	Black and Najafi (2024)	10.1007/s00158-024-03821-y	include	speed	p	P	p	P	agreed
P00837	Chimisso (2020)		exclude	none					agreed
P00842	Doosti et al. (2021)	10.1145/3485114.3485124	include	quality	p	–	–	p	adjudicated
P00860	Garrelts et al. (2021)	10.1016/j.procir.2021.11.221	include	none	–	–	–	–	agreed
P00875	Ma et al. (2023)	10.1002/nme.7221	include	none	–	–	–	–	agreed
P00890	Bujny et al. (2023)	10.1038/s41598-023-42009-0	include	speed	p	P	p	p	adjudicated
P00893	Yu (2024)	10.1093/jcde/qwae037	exclude	none					adjudicated
P00918	Martínez-Martínez et al. (2025)	10.1016/j.cma.2025.118044	include	speed	p	P	p	p	adjudicated
P00925	Jang and Kang (2020)	10.48550/arXiv.2008.07119	exclude	none					agreed
P00983	Xing et al. (2025)	10.1016/j.tws.2025.112942	include	none	–	–	–	–	agreed
P01004	Cang et al. (2018)	10.48550/arXiv.1807.10787	exclude	none					agreed
P01016	Ghansiyal et al. (2024)	10.1016/j.procir.2024.08.033	include	none	–	–	–	–	agreed

Record	Article	DOI	Decision	Stratum	R1	R2	R3	R4	Stage
P01105	Zhu et al. (2021)	10.1016/j.matdes.2021.110083	include	quality	p	–	–	p	adjudicated
P01151	Honarvar et al. (2026)	10.1016/j.cma.2026.119038	include	none	–	–	–	–	agreed
P01152	Abueidda et al. (2020)	10.48550/arXiv.2002.01896	exclude	none					agreed
P01164	Avdonyushkin et al. (2023)	10.15593/perm.mech/2023.3.01	exclude	none					adjudicated
P01176	Guo and Wang (2024)	10.31449/inf.v48i5.5295	exclude	none					adjudicated
P01214	Xiang et al. (2025)	10.1201/9781003658641-176	exclude	none					agreed
P01217	Parida and Dash (2023)	10.5281/zenodo.8003313	include	speed	F	p	p	p	adjudicated
P01219	Cheng et al. (2024)	10.1201/9781003483755-246	include	quality	p	–	–	P	adjudicated
P01232	Zhang et al. (2024)	10.1088/1742-6596/2791/1/012064	include	quality	p	–	–	F	adjudicated
P01268	Aroyan et al. (2026)	10.1051/matecconf/202642203010	include	none	–	–	–	–	agreed
P01289	Vulimiri et al. (2026)	10.1016/j.addma.2026.105094	include	none	–	–	–	–	agreed
P01320	Ouchaoui et al. (2024)	10.21494/iste.op.2024.1231	include	quality	p	–	–	p	adjudicated
P01390	Venugopal and Anand (2023)	10.1016/j.mfglet.2023.08.030	include	none	–	–	–	–	agreed
P01468	Behzadi and Ilieş (2021)	10.48550/arXiv.2105.03045	exclude	none					adjudicated
P01517	Li and Ye (2025)	10.32604/cmes.2025.072447	include	speed	P	p	–	P	adjudicated
P01670	Hosoi et al. (2025)	10.5281/zenodo.18597653	exclude	none					agreed
P01788	Torzoni et al. (2025)		exclude	none					agreed
P01813	Fernando et al. (2026)	10.1016/j.tws.2026.115420	include	quality	p	–	–	F	adjudicated
P01827	Carson and Hirschfeld (2026)	10.70671/dq9am588	include	none	–	–	–	–	agreed
P01870	Chu et al. (2024)	10.1016/j.matdes.2024.113087	include	none	–	–	–	–	agreed
P01900 (COI)	Drake et al. (2026)	10.1115/1.4071440	exclude	none					adjudicated
P01905	Chen and Lau (2026)	10.1039/d6mh00986g	exclude	none					adjudicated
P01907	Ignuta-Ciuncanu et al. (2025)	10.1016/j.icheatmasstransfer.2025.109098	include	none	–	–	–	–	agreed
P01960	N et al. (2025)	10.1016/j.mex.2025.103539	include	quality	F	–	–	p	adjudicated
P01971	Tskhondia (2025)	10.70792/jngr5.0.v1i3.95	include	quality	p	P	–	p	adjudicated
P01974	Ali (2026)	10.3390/jmmp10040117	include	speed	F	p	–	p	adjudicated
P01994	Tskhondia (2025)	10.13140/rg.2.2.16548.95369	include	none	–	–	–	–	adjudicated
P02042	Feng et al. (2026)	10.3390/app16094080	include	quality	p	–	–	p	adjudicated
P02074	Bhagat et al. (2026)	10.1007/s41024-026-00833-7	include	quality	p	–	–	F	adjudicated
P02084	Ignuta-Ciuncanu et al. (2026)	10.1016/j.ijheatmasstransfer.2026.129081	include	none	–	–	–	–	agreed
P02128	Jadhav et al. (2024)	10.1002/adfm.202404165	include	none	–	–	–	–	agreed
P02187	Erzmann and Dittmer (2024)	10.1093/jcde/qwae039	include	speed	p	p	p	P	adjudicated
P02296	Liu et al. (2026)	10.1145/3811393	include	none	–	–	–	–	agreed
P02306	Vlassis and Sun (2023)	10.1016/j.cma.2023.116126	exclude	none					agreed
P02328	Da and Chen (2023)	10.1016/j.ijheatmasstransfer.2023.124823	exclude	none					agreed
P02361	Krischer et al. (2024)	10.1007/s00158-024-03836-5	include	quality	p	–	–	P	adjudicated
P02476	Wang et al. (2020)	10.1016/j.cma.2020.113377	include	quality	p	–	–	p	adjudicated
P02479	Garland et al. (2021)	10.1016/j.matdes.2021.109632	include	none	–	–	–	–	agreed

2. Verdicts of the included articles

P00003 · Nguyen et al. (2026)

Adversarial Agents on Topology Optimization: Understanding the Fragility and Robustness of Deep Learning-based and Physics-Based Design Models under Adversarial Perturbation. arXiv preprint (cs.LG); no journal reference on the arXiv record. DOI 10.48550/arXiv.2608.22606. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Under bounded perturbations ($\varepsilon = 0.10$) of the initial-density input, the most fragile surrogate's compliance rises by a median 132.65× over its clean prediction, while SIMP re-run from the same attacked initializations returns a median 1.000× of the SIMP reference. **Location:** Sec. 6 Conclusion, pp. 24-25 (first quantitative ML-vs-classical comparison; the abstract states the same contrast without numbers, §5.1 rule 2); numbers from Table 1 (p. 18) and Table 3 (p. 23)

1. **R1: pass** (R1-matched). Tables 1 and 3, Sec. 5.4: surrogate predictions and SIMP re-runs start from the same attacked initialization with the same load, supports, volume target and solver settings, and both are scored by FE compliance against the same SIMP reference over the same 45 test problems. No speed claim is made. The surrogate threshold value and the achieved volumes are not stated.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) gives no baseline figures and names the same classical SIMP optimizer that Sec. 5.4 and Table 3 (p. 23) re-run, so (b) is not observed. No run time is reported for any method (Sec. 4-5); SIMP is the Top88 code on a 20×60 grid (Appendix A.5).

3. **R3: na** (R3-na-no-speed-claim). Speed appears only as motivation (abstract; Sec. 1). For C3: neither the cost of the 300 SIMP samples nor any training time is reported (Sec. 4).

4. **R4: partial** (R4-no-volume-check). Sec. 3.4 (p. 13), Tables 1-3, A.2 and A.3: compliance of every returned (thresholded) surrogate design and every SIMP re-run is computed by FE under the original problem for all 45 test problems, and failures are counted by exceedance thresholds, with disconnected designs treated as near-singular failure indicators (Sec. 3.4, 5.2). Neither the threshold value nor the volume fraction of the thresholded predictions is reported anywhere in the article, so volume-constraint satisfaction of the returned designs is unreported.

P00006 · Joo et al. (2026)

Explainable quantum neural networks for multi-material topology optimization. arXiv preprint (cs.CE). DOI 10.48550/arXiv.2607.00438. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On four unseen 2D load cases (50×25), XQNN material labels predicted from early-stopped MMA states matched converged MMA layouts at 76.7–93.5% accuracy, with compliance 0.96–1.16 times the MMA reference across 2–4 materials. **Location:** Sec. 2.3 and Fig. 3a; per-case values printed in Supplementary Figs. S6-S8. Chosen under §5.1 rule 4 as the emphasized test: the abstract, introduction and discussion (the article has no separate conclusion) compare with MMA only qualitatively, and the first number in the main text is the validation accuracy of Sec. 2.2

1. **R1: partial** (R1-loose). Sec. 2.3, Figs. 3 and S6-S8: XQNN layouts and the converged MMA references share the mesh, loads and material settings, and compliance is reported as C_pred/C_GT for every case. Matching is loose: XQNN assigns labels element by element and does not enforce the volume constraints at inference (Sec. 2.3), no achieved volume is reported, so ratios below 1 (3D means 0.83-0.98) may reflect extra material [inferred]; no compute budget or iteration count is compared.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no classical baseline and gives no numbers, so it cannot report a weaker baseline than Sec. 2.3 and Figs. S6-S8; (b) is not observed. No run time or iteration count of any method is reported (Sec. 2-4, Supplementary Notes 1-3). The comparator is the authors' adaptation of a published MMTO MATLAB code (Zheng et al. 2024), run to a grayscale ratio below 0.02 (0.01 for HR) (Methods 4.1).

3. **R3: na** (R3-na-no-speed-claim). No quantified speed claim; the early-stopping acceleration is asserted only qualitatively (Sec. 2.2; Supplementary Note 2; Table S6). For C3: training uses the MMA histories of three 50×25 problems per material setting and 100 Adam epochs (Supplementary Note 3); neither data-generation nor training time is reported, and every inference still needs an MMA run up to kpd (Sec. 2.1).

4. **R4: partial** (R4-no-constraint-report). Sec. 2.3, Figs. 3 and S6-S8: the normalized compliance of each returned layout is reported against the MMA reference [FE evaluation inferred; the procedure is not described], and all OOD, HR and 3D cases are shown, so no selection issue. Volume-constraint satisfaction is not reported, and the authors state it is not enforced at inference. Weakly connected or disconnected parts in the two-material HR cases (C = 2.68 and 2.96) are described (Sec. 2.3) but not counted as failures.

P00019 · Sun et al. (2026)

Multi-material Multi-physics Topology Optimization with Physics-informed Gaussian Process Priors. arXiv preprint (cs.LG). Version of record: Computer Methods in Applied Mechanics and Engineering 461, 119163

(Nov 2026), doi 10.1016/j.cma.2026.119163 (Crossref title search, same four authors); not open access (Crossref licence metadata lists only Elsevier TDM licences and an author-manuscript licence from 18 Jun 2027; Elsevier API openaccess flag 0), so not read. DOI 10.48550/arXiv.2602.17783. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). With a mass constraint on four 2D multi-material compliance benchmarks, PIGP's median compliance over 10 runs is slightly above ordered SIMP's and PolyMat's (e.g., MBB 1.049 vs 0.982 and 0.987); with a cost-only constraint it matches PolyMat. **Location:** Section 4.1, Table 4 (first quantitative classical comparison; the abstract, introduction and conclusion compare with the open-source codes and COMSOL only qualitatively)

1. **R1: partial** (R1-loose). Sec. 4, 4.1, Tables 2-4: every method is scored by FEM at the same mass or cost target, and achieved values are reported (e.g., MBB mass 6006.3 PIGP, 5990.8 ordered SIMP, 5988.0 PolyMat; target 6000 [derived]). Matching is loose: discretizations differ (Table 3), length-scale control differs (filter radius 6 and 2 mm vs the PGCAN resolution, which Sec. 4.2 links to feature complexity), and PIGP runs up to 10,000 epochs while the iteration counts and stopping rules of ordered SIMP and PolyMat are not stated. No runtimes are compared.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) gives no numbers and names the same comparators that Sec. 4 runs (open-source codes and COMSOL), so it does not report a weaker baseline than Table 4 or the other results; (b) is not observed. No runtime of any method is reported (Sec. 4-5). For the record: classical codes run on an Intel i7-11700K CPU and PIGP on an NVIDIA A100 GPU (Sec. 4), with no speed claim tied to this asymmetry.

3. **R3: na** (R3-na-quality-stratum). Quality stratum with per-problem physics-informed training and no offline data (Sec. 3); no speed claim. For C3: no runtime, training time or GPU-hours are reported for any method; the conclusion says only that PIGP costs more to train than SIMP.

4. **R4: partial** (R4-no-failure-rate). Sec. 4 opening and the captions of Tables 4, 5, 8, 11 and 15: the objectives of the final topologies of all methods are evaluated by FEM (ABAQUS after binarization in Sec. 4.4), achieved mass and cost values are reported, and statistics cover all 10 runs (medians in Table 4). Defective designs are not counted: gray areas (Sec. 4.3) and thin or disconnected features (Sec. 4.4) are described only qualitatively, and whether the Table 4 designs were binarized before the FEM evaluation is not stated.

P00029 · Singh et al. (2025)

Accelerating metamaterial topology optimization using deep super-resolution networks. arXiv preprint (physics.comp-ph). DOI 10.48550/arXiv.2511.04795. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). After training, running SIMP at 48×48 and upscaling with the EDSR network gives 192×192 unit cells in 24-28 s on average, versus 386-493 s for SIMP run directly at 192×192 (5-7% of the cost), with mean objective errors of 1.57-4.24%. **Location:** Abstract (quantitative comparison with the classical method, §5.1 rule 1); numbers in Sec. 4.1-4.4, Tables 1, 3, 5, 7 (times) and Tables 2, 4, 6, 8 (errors)

1. **R1: partial** (R1-loose). Tables 1-8, Sec. 3.4 and 4: average times are compared with SIMP runs at the same 192×192 resolution, and quality is reported for the same 150 test designs (objective error 1.57-4.24%, volume-fraction error 1.04-1.95%). Matching is loose: the SIMP stopping rule and iteration count are not stated, volumes differ by 1-2% on average, the sign of the objective error is not given, and no equal-quality or equal-budget run is attempted. A verifier who follows P00176 (mean compliance error about 4%, no matching) may choose fail (R1-unmatched).

2. **R2: pass** (R2-none-established). Sec. 4: the comparator is the group's published 55-line FEniCS SIMP code on a Xeon 6230 workstation; the device per method is not stated. At 192×192 (≈7.4×10^4 DOFs [derived]) marker (c) does not apply; (b) is not observed (the abstract's 5-7% matches Tables 1-7); (d) and (e) cannot be derived without an iteration count. A stronger classical comparator would start the 192×192 SIMP run from the upsampled 48×48 design that the pipeline already computes [inferred]; it is not tried. Unsure, so pass.

3. **R3: fail** (R3-no-cost). Sec. 3 (end of the overview) and abstract: the speed claim applies once the network is trained, and sample preparation and training are called a one-time offline expense, but

neither is quantified and no break-even is given. For C3: 1000 samples per objective, each with a 192×192 and a 48×48 SIMP run (Sec. 3.1) [inferred ≈4.8×10^5 s per objective at the bulk-case times]; the saving is 433 s per design (457 - 24 s), so break-even is about 1,100 uses per objective before training cost [derived].

4. **R4: partial** (R4-no-failure-rate). Sec. 3.4 and Tables 2, 4, 6, 8: the objective of each returned (thresholded) design is recomputed by converting the image to a mesh and re-analysing it, and the volume-fraction error is reported, averaged over all 150 test samples per objective; figures show four examples per objective, but the numbers cover the test sets. Disconnected, non-periodic or otherwise failed designs are neither defined nor counted.

P00051 · Radler et al. (2025)

Diverse Topology Optimization using Modulated Neural Fields. arXiv preprint (cs.LG); v1 was titled 'Generative Topology Optimization: Exploring Diverse Solutions in Structural Design' (method name GenTO); no journal reference on the abs page. DOI 10.48550/arXiv.2502.13174. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). TOM's sampled designs come close to, but average above, classical compliance: 0.75±0.03 vs 0.68 (SIMP) and 0.67 (deflated barrier) on MBB, 0.69±0.13 vs 0.59/0.58 on the cantilever, and 1.31 vs 0.99 ($\times 10^{-3}$, SIMP) on the 3D bracket. **Location:** Sec. 4.4 and Table 1, p. 9: first quantitative classical comparison. Abstract, introduction and conclusion are qualitative. Chosen by §5.1 rule 4: near-optimality is claimed in the abstract and introduction, and the conclusion stresses adherence to mechanical requirements, while the speed claim appears only in the introduction and one results heading

1. **R1: partial** (R1-loose). Stratum Q; R1 applies as written (PROTOCOL §11). Sec. 4.3 and Tables 1-2 (pp. 9, 14): all designs are scored on the same high-resolution mesh. Budgets and outputs differ: TOM uses 9,000-25,000 solver calls vs 400 (FeniTop) and 3,190-4,922 (DB), and samples a continuous family vs one or two designs. On MBB, TOM's mean volume exceeds the 53.5% target (55.08%) while its compliance is still higher. The FeniTop stopping rule is a fixed 400 iterations; DB's is not stated.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The v2 abstract (p. 1) gives no numbers and no speed claim, so it cannot report a weaker baseline than Table 1; (b) is not observed. For the record, Sec. 4.4 attributes part of DB's run time to its public implementation's mesh refinement, which bears only on the speed alternative.

3. **R3: na** (R3-na-per-problem). TOM is data-free and trains one network per problem with the solver in the loop (Sec. 3.1), so its cost sits inside R1/R2; the reported TOM times include that training (Table 1).

4. **R4: partial** (R4-floaters-not-counted). Sec. 4.3 and Table 1: compliance and volume of the returned designs come from a FEniCSx analysis on a common mesh. Load-violating designs are counted and filtered (TOM 8% on the cantilever). Floaters (disconnected pieces) are acknowledged only qualitatively (Sec. 4.4; post-processing shown in App. D.3 but not used for Table 1 [inferred]). The number of TOM samples behind the statistics is not stated.

P00080 · Padhy et al. (2024)

VoroTO: Multiscale Topology Optimization of Voronoi Structures using Surrogate Neural Networks. arXiv preprint (cs.CE; also math.NA), CC BY 4.0; comment 'Submitted to Engineering with Computers'. Version of record: Engineering with Computers 41(2):1123-1139, doi 10.1007/s00366-024-02077-w, online 2024-10-29 (Crossref, OpenAlex); closed access per OpenAlex, and Crossref lists only Springer TDM licences; not read. Crossref also links a Research Square preprint (doi 10.21203/rs.3.rs-4339435/v1), which is not in the pool. DOI 10.1007/s00366-024-02077-w. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). For the 40x20 mid-cantilever, NN-surrogate multiscale Voronoi TO takes 0.24 min after a one-time 28.4 min (data plus training), versus a projected 528 min for concurrent homogenization-based multiscale TO and 4,986 min for brute-force fine-scale TO. **Location:** Sec. 3.3 ('Is Offline Computation Justified?', pp. 10-11). The abstract and conclusion state the advantage over concurrent homogenization only qualitatively, and the introduction has no classical comparison. The main text's first classical comparisons are compliance values (Sec. 3.2.1-3.2.2); rule 4 takes Sec. 3.3, the comparison the conclusion emphasises

1. **R1: partial** (R1-loose). Sec. 3.3 (pp. 10-11): VoroTO's 0.24 min (plus the one-time 28.4 min) is set against projected times for concurrent homogenization (528 min) and brute-force fine-scale TO (4,986 min). Neither classical scenario was run, so no classical design exists at the timed budget. Quality is still reported for the same mid-cantilever: the VoroTO design (compliance 68.3 as predicted; 74.6 at volume 0.65 against the 0.6 target on FE re-analysis) is set against the classical single-scale 88-line design (61.05) and the infill design of [96] (76.86) (Sec. 3.2.2, pp. 8-9), and the claim is paired with a surrogate error below 10% against exact homogenization (Sec. 3.3, p. 11). Matching is loose: the volume is not matched, and the projections assume 300 iterations with no quality target. Neither fail marker is established: a quality comparison exists, and the projected classical runs have no output that could be shown better.

2. **R2: partial** (R2-inefficient-impl [inferred]). Sec. 3.3: both classical times are projections. The concurrent-homogenization estimate assumes serial homogenization of all 800 elements at each of 300 iterations on a laptop, with no parallel or reuse strategy. The brute-force estimate multiplies one 16.62 min iteration at 1600 x 800 (about 2.56 million DOFs [derived]) by 300, with the solver not stated; that per-iteration time looks slow for the size [inferred] and is an input for an R2(e) re-test. (b) and (c) do not apply (one laptop for all runs). (d) is not established: 300 is VoroTO's own iteration cap, and the article gives no classical convergence data.

3. **R3: pass** (R3-full-cost). Sec. 3.3: data generation (26.4 min) and training (2 min) are reported and set, with the 0.24 min run, against one projected classical run. The total, 28.64 min [derived], is below either projection, so the method breaks even at the first use.

4. **R4: partial** (R4-no-failure-rate). Sec. 3.2.2 and 3.2.4: for the mid-cantilever and heel-bone designs, the microstructures were rebuilt and re-analysed by homogenization FEA and macroscale FEA, and the re-analysed volume is reported (0.65 vs target 0.6). Figs. 14 and 17-19 report optimizer-side compliance values [inferred: NN-predicted]. Disconnected or failed microstructure layouts are not counted; filtering promotes connectivity, but connectivity is not measured.

P00081 · Chadha and Kumar (2024)

PyTOaCNN: Topology optimization using an adaptive convolutional neural network in Python. arXiv preprint (cs.CE; 24 pages). Version of record: Soft Computing 30(1) 653-673 (online 25 Nov 2025), doi 10.1007/s00500-025-10919-y (Crossref); the Springer page marks it as not open access, so it was not read. Its abstract is reworded, e.g. it says the results hold for test problems and puts the code in Appendix A. DOI 10.48550/arXiv.2404.12244. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). For the mid-load beam on a 100×100 grid, CNN designs at seven prescribed volume fractions differ from top88 SIMP designs by 0.053–5.4% in volume and 0.041–2.66% in compliance (Table 1); the authors call such errors negligible. **Location:** Section 7.1.1 and Table 1 (first quantitative classical comparison, in the main text; the Abstract, Section 8 and Introduction compare only qualitatively)

1. **R1: partial** (R1-loose). Stratum Q; R1 applies as written (PROTOCOL §11). Sec. 7 and Table 1 (p. 12): CNN and MATLAB designs share the problem, the grid and the prescribed volume fraction, and the CNN output is scored in the same MATLAB code. Matching is loose: the CNN's own volume misses the target by up to 5.4% (Table 1; up to 14.4% in Table 4), and the objective error is not corrected for that volume difference. The stopping rule of top88 and topX is not stated in the article (TOPress: maxit 100, Sec. 5), and no compute budget is compared.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names the same open-source TO codes that Sec. 7 uses (top88, TOPress, topX) and gives no numbers, so it does not report a weaker baseline than Tables 1-7; (b) is not observed. No runtime of any method is reported anywhere.

3. **R3: na** (R3-na-no-speed-claim). Quality stratum: no quantified speed claim against a classical method. For C3: data = 95 MATLAB TO runs per problem (68 for topX) and training = 2000 epochs; neither time nor hardware is reported (Sec. 5, 6.5).

4. **R4: partial** (R4-no-failure-rate). Sec. 7 (p. 13) and Tables 1-7: the returned grayscale CNN field is evaluated in the same MATLAB code, so the objective is FE-computed, and the volume error is reported for every case. Failed or disconnected outputs are neither defined nor counted. The authors

say they picked the volume fractions of Tables 1 and 2 arbitrarily from the training set (Sec. 7.1.2, p. 16); in Tables 1, 2, 5, 6 and 7 all of them lie inside the training sets [Tables 5-7 derived from the training ranges in Sec. 5, 7.1.3, 7.1.4 and 7.2].

P00092 · Zhong et al. (2023)

Topology optimization with text-guided stylization. Structural and Multidisciplinary Optimization 66:256 (2023), published online 12 Dec 2023; open access, CC BY 4.0 (article footer). Preprint: arXiv 2310.15506 v1 (24 Oct 2023, CC BY 4.0). DOI 10.1007/s00158-023-03686-7. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). With only the mechanical loss active, the hash-encoded neural field reaches nearly the same compliance as SIMP on three 2D benchmarks at matched volume. Bridge: 181.08 vs 181.06; MBB: 40.80 vs 39.48; L-bracket: 177.23 vs 169.80. **Location:** Sec. 4.1 'Mechanical performance', p. 10, and Fig. 6 (first quantitative classical comparison; the abstract, conclusion and introduction hold no quantitative classical comparison, §5.1 rule 2)

1. **R1: partial** (R1-loose). Sec. 4.1, p. 10: SIMP uses the same penalty (p = 2.0), the filter radius was tuned, and the achieved volumes match within 0.002. Matching is loose: SIMP's iteration count, stopping rule and FE resolution are not stated, while the proposed method stopped at 100 iterations. The target volume fraction of each task is not given, and one run per method and task is compared, although the method varies by about ±3% across random initializations (Sec. 4).

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no baseline and gives no numbers, so it cannot report a weaker baseline than Sec. 4.1; (b) is not observed. No SIMP runtime is reported anywhere; SIMP is the public DTU Python code with a tuned filter radius (Sec. 4.1, pp. 9-10).

3. **R3: na** (R3-na-quality-stratum). Primary outcome is quality only, and the method learns per problem with no offline training or dataset (Sec. 4, 'Gradients computation'). For C3: 0.587 s per iteration at 256x256 on an RTX 4080 laptop GPU (Sec. 4.1, Fig. 11); 100-500 iterations per design.

4. **R4: partial** (R4-no-failure-rate). Sec. 4.1 (p. 10) and Fig. 6: compliance and volume are reported for SIMP and both variants of the method on all three tasks, from FE analysis (the method's own FE on the pooled 64x64 field [inferred]), and the Lmech-only volumes match SIMP's within 0.002. Failed designs are neither defined nor counted: the Lmech-only variant has no connectivity check, the target volume fraction is not given although the authors state that their penalty formulation misses it (Sec. 3.1, p. 6; Sec. 4.1, p. 10), and one run per task is shown from a method with about ±3% run-to-run spread (Sec. 4, p. 9) with no stated rule for which run is reported.

P00136 · Gladstone et al. (2021)

Robust Topology Optimization Using Multi-Fidelity Variational Autoencoders. arXiv preprint (cs.LG); no journal reference on arXiv. DOI 10.48550/arXiv.2107.10661. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On a 100×100 L-bracket with uncertain load angle, gradient search in a VAE latent space with a neural robust-compliance surrogate found a design with robust compliance 69.84, below the best of 1,000 deterministic SIMP designs (71.59). **Location:** Sec. 4.1.3 and Fig. 11 (first quantitative comparison with a classical method; the Abstract, Introduction and Sec. 5 have no quantitative one); key phrase from the Abstract

1. **R1: partial** (R1-loose). Sec. 4.1.3, Fig. 11: the comparator is the best of the 1,000 deterministic SIMP designs, which are also the method's training pool, so the ML pipeline's budget includes the comparator's full cost plus 7-point evaluation of every design and network training. No equal-budget or equal-time comparison is made, and the returned design's volume fraction is not reported (for generated samples, under 1% exceed 0.4 and the maximum is below 0.405, Sec. 4.1.1).

2. **R2: fail** (R2(b)). Stratum Q; R2 applies through criterion (b) (PROTOCOL §11, 2026-09-24). The abstract (p. 1) reports only the comparison with the training-set designs (better robust compliance) and mentions standard RTO only for its number of FE evaluations. Sec. 4.1.3 (p. 14) states that the ML design cannot outperform the best robust design obtained from SIMP, and Sec. 5 (p. 21)

states that it could not be shown better than a finite-element approach. The stronger classical baseline appears only in the results, and without a value.

3. **R3: na** (R3-na-quality-stratum). Quality-stratum primary outcome. For C3: data generation is reported (3.55 h for 600 SIMP designs, about 21 s each [derived]) and training only per iteration; no classical robust-TO time and no break-even, although the abstract claims computational efficiency over standard RTO (Secs. 4.1.4 and 4.2).

4. **R4: partial** (R4-no-constraint-report). Sec. 4.1.3, Figs. 11 and 17: robust compliance of the returned designs is reported with the exact-value notation Qrob, not the surrogate QNN [FE evaluation inferred]. The returned designs' volume fractions are not reported, Sec. 5 lists checking the structural integrity of the optimized designs as future work, and no failure rate is given.

P00140 · Elingaard et al. (2021)

De-homogenization using Convolutional Neural Networks. arXiv preprint (cs.LG). Version of record: Computer Methods in Applied Mechanics and Engineering 388 (2022) 114197, doi 10.1016/j.cma.2021.114197, published online 2021-10-21, open access under CC BY (OpenAlex: hybrid; DTU Orbit lists the 'Final published version'). Neither copy could be read by script: ScienceDirect serves a captcha and the DTU Orbit PDF sits behind a Cloudflare challenge (HTTP 403); neither was bypassed. DOI 10.1016/j.cma.2021.114197. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). A CNN replaces the least-squares step of classical de-homogenization. The resulting single-scale designs come within 7-25% of the homogenization-based reference performance, at a small fraction of the cost. The body quantifies the saving as about 5-10x over least-squares de-homogenization. **Location:** Abstract (first quantitative comparison with a classical method; §5.1 rule 1). The cost part is qualitative in the abstract and is quantified in Sec. 4.2.1 (p. 18) and the Conclusions (p. 22) as a factor of 5 to 10 over state-of-the-art de-homogenization

1. **R1: partial** (R1-loose). Abstract, Sec. 4.2.1 (pp. 17-18) and Conclusions (p. 22): the 5-10x factor compares this work's CPU times (about 1.3 s and 2.2 s) with a time cited from [16] (about 10 s, 80x40 -> 1600x800) on another laptop, which the authors call not directly comparable. Quality is reported only against the homogenization reference (performance ratio 1.0677-1.5260 over 30 cases [derived]; 7-25% in the abstract for suitable parameters), and the authors present that loss as a trade-off for cost (p. 19). The quality of [16]'s designs is not reported and no matching of quality or budget is attempted, so matching is loose (§5.3 R1 partial). Fail would apply if [16]'s designs were shown to be measurably better. The alternative top88 comparison (four orders of magnitude) produces no top88 design and would fail R1 (R1-no-quality) if taken as primary.

2. **R2: pass** (R2-none-established). Sec. 4.1 and 4.2.1: both sides are timed on laptops (CNN on a single Intel Core i7 2.6 GHz CPU; [16] on a laptop of unstated model and device), and the ML side runs on a CPU, so (c) does not apply. The abstract gives no baseline figure (b). No iteration count exists for [16], so (d) and (e) cannot be derived. The authors flag that the cross-paper times are not directly comparable. Unsure, so pass. Pointer for the top88 alternative: 16 s and 87 s per iteration at about 2.1 and 5.8 million DOFs [derived] can be checked for (e) with the project's baseline at the same size; the 1000 iterations are assumed, not run.

3. **R3: partial** (R3-no-data-cost; R3-no-amortization). Sec. 4.1: training time (about 1 h on a Titan X for 15 epochs) is reported. The cost of generating the 20,000 synthetic orientation fields is not stated (it needs no TO runs; small [inferred]). No break-even or amortization is given, and the claim is not explicitly limited to per-query use. For C3 [derived, cross-hardware, rough]: 3,600 s of training over a saving of about 8.7 s per use vs [16] gives about 400 uses; vs the top88 alternative, fewer than one use.

4. **R4: partial** (R4-no-failure-rate). Tables 2-5 and App. A Table 1: compliance and volume fraction are reported for the post-processed design in every case (all 30 cases, no selection); FE evaluation is inferred, as the method is not described. Volume overshoot is visible per case and discussed (Sec. 4.2-4.3). Non-load-carrying boundary artefacts appear in many designs (Sec. 4.2, p. 16) but are not counted, and no disconnection or failure count is given.

P00141 · Behzadi and Ilieş (2021)

GANTL: Towards Practical and Real-Time Topology Optimization with Conditional GANs and Transfer Learning. arXiv preprint (cs.LG; also cs.AI). Version of record: Journal of Mechanical Design 144(2) 021711, doi 10.1115/1.4052757, retitled 'GANTL: Toward Practical and Real-Time Topology Optimization With Conditional Generative Adversarial Networks and Transfer Learning' (doi.org resolves to the ASME article page; volume, issue and article number taken from the ASME URL in a web-search listing, same article id 1121902). The ASME page returned HTTP 403 to scripted access, so its open-access status and content were not checked. DOI 10.48550/arXiv.2105.03045. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). For seen boundary conditions with one random load, GANTL predictions at six resolutions (40×80 to 200×400) differ from the SIMP designs by 1.00% average compliance error (std 2.5%), 2.12% MSE and 96.72% binary accuracy. **Location:** Sec. 4.2 para 3 and Table 2 (first quantitative comparison with a classical method; the abstract, conclusion and introduction hold none)

1. **R1: partial** (R1-loose). Stratum Q; R1 applies as written (PROTOCOL §11). Tables 2-5, Sec. 3.4-3.5, 4.2: predictions are compared with SIMP designs for the same loads, BCs and prescribed volume fraction 0.5, but the achieved volume fraction of the predictions is not reported, and the compliance error takes the absolute value of the summed difference (Sec. 3.5, p. 6), so its sign is hidden. The SIMP penalty value and stopping rule are not stated, and no run time exists on either side, so no budget is matched. No speed claim is quantified.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no classical baseline and gives no numbers (its comparisons are with other deep-learning networks), so it cannot report a weaker baseline than Tables 2-5; (b) is not observed. The SIMP references come from the public 88-line MATLAB code [4] (Sec. 3.4, 4.1); the cited 1.85 s per iteration (Sec. 1) is not used in any comparison.

3. **R3: na** (R3-na-quality-stratum). No quantified speed claim against a classical method. For C3: 11,000 + 1,500 SIMP training cases (Sec. 3.4, 4.1); no data-generation, training or inference time is reported; the conclusion refers the data-generation cost to [5] (Sec. 5).

4. **R4: partial** (R4-no-constraint-report). Sec. 3.5 (p. 6) and Tables 2-5: the compliance error is the absolute difference of the summed element compliances of prediction and SIMP design over the SIMP total, i.e. a relative total-compliance error [derived from the formula], reported for every test set (250-7,473 cases per row) with MSE and binary accuracy; compliance of predictions presumably from FE analysis [inferred; method not stated]. The volume fraction of predictions is not reported, and disconnected predictions are acknowledged (Sec. 4.3) but not counted. The topology-loss improvement (Fig. 11) rests on examples shown without a stated selection rule.

P00148 · Deng and To (2021)

A Parametric Level Set Method for Topology Optimization based on Deep Neural Network (DNN). arXiv preprint (math.OC). Version of record: 'A Parametric Level Set Method for Topology Optimization Based on Deep Neural Network', Journal of Mechanical Design 143(9) 091702 (published online 2021-03-18), doi 10.1115/1.4050105 (pool P00544; ASME licence per Crossref, closed per OpenAlex); not accessed. DOI 10.48550/arXiv.2101.03286. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). MBB beam (200x100, volume fraction 0.4): the DNN level-set designs reach compliance 33.39 (one hidden layer) and 32.57 (two layers), versus 33.48 for the SIMP benchmark from the 88-line code; the authors call them quite close. **Location:** Sec. 4.1, Figs. 8 and 11 (first quantitative comparison with a classical method; the abstract, introduction and conclusion make only qualitative claims of diverse and competitive designs)

1. **R1: partial** (R1-loose). Sec. 4.1-4.2 (pp. 8-15), Figs. 8, 11, 15, 18: compliance of the DNN level-set designs is set against one SIMP (88-line code, filter radius 2) design per problem. Matching is loose: the DNN runs use a fixed Lagrange multiplier (5 for the MBB beam, p. 8; 3 for the cantilever, p. 12) rather than an enforced constraint, and no achieved volume fraction is reported for either method; the SIMP penalty, optimizer and stopping rule are not stated; the level-set compliance uses an ersatz material and a smoothed Heaviside, while the SIMP compliance includes grey elements [inferred]. Quality is reported but matching is loose.

2. **R2: na** (R2-na-quality-only). Stratum Q: no speed or cost comparison, and no run time of any method is reported, so criteria (a)-(e) do not apply.

3. **R3: na** (R3-na-per-problem). Per-problem neural reparameterization with no offline training data (Sec. 2-3), and no speed claim.

4. **R4: partial** (R4-no-constraint-report). Figs. 8, 12, 15, 19: compliance of each returned design comes from the FE analysis inside the optimization [inferred], and every architecture tried is reported with its compliance (no selection rule issue). Volume-constraint satisfaction is not reported for any design, although the volume is handled with a fixed Lagrange multiplier; no failed or disconnected designs are counted.

P00149 · Zheng et al. (2021)

Data-driven topology optimization of spinodoid metamaterials with seamlessly tunable anisotropy. Computer Methods in Applied Mechanics and Engineering 383, 113894. DOI 10.1016/j.cma.2021.113894. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The trained DNN surrogate returns a spinodoid RVE's homogenized stiffness in about 0.001 s per sample, versus roughly 5 min for FE homogenization, so stiffness evaluations inside the two-scale optimization become orders of magnitude cheaper. **Location:** Section 4.1, p. 11 (first comparison with a classical method in the main text; abstract, conclusion and introduction hold only qualitative claims); numbers in Appendix B, Table B.1, p. 23

1. **R1: pass** (R1-matched-per-evaluation). The primary outcome is a per-evaluation comparison (Section 4.1, p. 11; Table B.1, p. 23), and at that level the DNN output is checked against the same FE homogenization it replaces: $R^2 \geq 0.996$ for every stiffness component on 2,130 held-out FEM samples, and $R^2 \geq 0.999$ for the AD sensitivities against central differences of FEM stiffnesses (Section 5.1, p. 12, Figs. 5-6). The speed comparison therefore sits within the ML method's reported error of the classical result. No fail condition holds: a quality comparison exists and the DNN output is not shown to be worse than its reported error. The unstated RVE mesh ('for the meshes used in our study', p. 11) and the authors' remark that the runtimes are only a qualitative measure (Appendix B, p. 23) concern the baseline's cost and the timing protocol, not quality matching. No FE^2-type optimization is run, so there is no design-level speed claim to match. Unsure → pass.

2. **R2: pass** (R2-none-established). Table B.1: FE homogenization uses an in-house C++ FEM code on 16 MPI cores (Euler IV), while the DNN runs on one CPU without parallelization. No GPU is used, so (c) does not apply; the abstract gives no baseline figure (b); no fixed iteration count applies (d). Solver type and RVE mesh are not stated; no inefficiency is evident. Unsure → pass.

3. **R3: partial** (R3-no-data-cost; R3-no-amortization). Table B.1 (p. 23): DNN training 10 min; FE homogenization about 5 min per sample. Section 5.1 (p. 12): 19,170 training + 2,130 test samples. The total data-generation cost is never stated [inferred ≈1,775 h of 16-core time], and data and training were reused from Kumar et al. 2020a (Table B.1 note). No break-even is given. Appendix B frames the gain per prediction, but the conclusion's speed-up statement is not limited to per-query use (Section 6, pp. 20-21).

4. **R4: partial** (R4-surrogate-predicted). Surrogate outputs are checked against FEM on 2,130 held-out samples (Section 5.1). The optimized compliances of the returned designs (Figs. 7c, 10c, 12c, 14c) are computed with DNN-predicted stiffness; no FE homogenization or fully resolved FE re-check of the final design parameters is reported [inferred from Sections 3–5 and Appendix A], and the achieved relative density is not reported. Partial because the macroscale FE is real and the surrogate is independently validated; a stricter reading (design performance rests on the surrogate) gives fail.

P00150 · Halle et al. (2021)

An Artificial Intelligence–Assisted Design Method for Topology Optimization without Pre-Optimized Training Data. Applied Sciences 11(19), 9041 (published 28 Sep 2021); open access, CC BY. DOI 10.3390/app11199041. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On 100 random 64×64 validation cases, the trained PEN predictor returns a design in about 7.3 ms on average versus 1.9 s for top88 SIMP, roughly 259 times faster, with

compliance described as almost the same. **Location:** Sec. 3.2 Results and Discussion, p. 16, Fig. 14a (first quantitative comparison with a classical method; Abstract, Conclusions and Introduction hold only qualitative comparisons)

1. **R1: partial** (R1-loose). Sec. 3.2 (p. 16), Fig. 14, Table 3 (p. 17), Fig. 15 (p. 18): PEN inference time is compared with complete top88 runs whose stopping rule is not stated (Table 2, p. 15, lists p, r_min, ν, E and the force only). Quality is reported: FE compliance for all 100 cases (scatter only) and six printed pairs, in which PEN is 1.8-5.6% worse in five and 10.2% better in one [derived]. Volume is matched only approximately (ΔM = |M_tar − M_is|, Eq. 21, p. 9, up to 0.010 in the six printed cases; the constraint is a penalty, p. 17). No attempt is made to match top88 time or iterations to PEN quality.
2. **R2: pass** (R2-none-established). Sec. 3.2 (p. 16) and Table 5 (p. 19): the timed classical method is top88 [29] at 64×64 ($\approx 8.4\times10^3$ DOFs [inferred]); arXiv v2 (p. 12) names the DTU TopOpt Python codes as the source of the top88 runs. The text establishes none of (a)-(e): the abstract gives no baseline figure (b); the problem is far below 10^5 DOFs, so (c) cannot apply even if PEN ran on the GPU [inferred]; no iteration count or per-iteration time is given, so (d) and (e) cannot be derived. The 88-line code is vectorized and uses a sparse direct solve [recalled from ref. 29; not described in the article], so no inefficiency is evident. The hardware of both timings is not stated (Table 1, p. 12, is the training configuration only); this is recorded as a reporting gap (hardware_per_method_stated = partial). Unsure → pass.
3. **R3: pass** (R3-full-accounting). Sec. 3.3, Table 5, p. 19: training time T_P = 3.25 h is reported and folded into a break-even count e_BEP = 6226 predictions. There is no separate data-generation cost, because training uses random inputs evaluated by FE inside the training loop (Secs. 1, 2.8). Not included: the cost of the hyperparameter grid search that compared predictions with conventional TO results (Sec. 3.1).
4. **R4: partial** (R4-no-failure-rate). Compliance of the returned PEN designs is computed by FE analysis (compliance evaluator, Sec. 2.3; Fig. 14b for all 100 cases). No post-processing is applied. Volume deviation ΔM is printed only for the six Fig. 15 examples (Table 3). No failure or infeasibility count is given, although the authors note that some geometries lack details (Sec. 3.2, Fig. 15 columns four and five).

P00161 · Nakamura and Suzuki (2020)

Deep learning-based topological optimization for representing a user-specified design area. arXiv preprint (cs.CE; also cs.LG). No journal version found (pool search and web search, 2026-09-24). DOI 10.48550/arXiv.2004.05461. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On 32x32 compliance problems with user-specified design areas, the CNN returns a design in 0.07634 s versus 0.4393 s for the 88-line SIMP code on the same CPU (83% less time); best-80% compliance error 2.75%. **Location:** Conclusion (Sec. 5, bullet list, p. 11), with Sec. 4.3 and Table 1 (p. 8) for the times and Table 2 (p. 9) for the errors. The abstract compares with the MATLAB code only qualitatively. The conclusion holds two quantitative classical comparisons (2.75% compliance error; 83% time cut); rule 4 takes the time cut, which the abstract's only classical comparison emphasises

1. **R1: fail** (R1-unmatched). Sec. 4.3, Table 1 (p. 8) and Table 2 (p. 9): CNN inference (0.07634 s per case) is set against complete SIMP runs (0.4393 s per case), while the CNN outputs are measurably worse than the SIMP designs: the mean volume-fraction error is 9.70%, so the volume is not matched, and the worst 20% of test outputs, which include disconnected designs (Fig. 13; Sec. 4.3.2, pp. 9-10), have a mean compliance error of 103,872%. The conclusion's claim of near-identity with SIMP rests on the best 80% only (2.75%; Sec. 5, p. 11). No matching, such as a SIMP run stopped at equal quality or equal time, is attempted.
2. **R2: pass** (R2-none-established). Sec. 4.2-4.3: the baseline is the published 88-line MATLAB code, which its cited paper presents as an efficient implementation, run on the same CPU as the CNN. (c) does not apply (2,178 DOFs [derived]; CPU for both). The SIMP iteration count and stopping rule are not stated, so (d) and (e) cannot be derived; (b) not observed. Unsure, so pass.
3. **R3: fail** (R3-no-cost). Sec. 4.2-4.3: the speed claim reports neither training time nor data-generation time; only the number of SIMP samples (370,000) is given. [derived] At 0.4393 s per SIMP run the data cost about 45.2 h and are recovered after about 448,000 uses at 0.363 s saved per use, before training.

4. **R4: partial** (R4-no-failure-rate). Sec. 4.3.2, Table 2, Figs. 13-14: compliance of CNN outputs is computed against SIMP [FE analysis inferred], volume-fraction error and design-area satisfaction are reported, and the best-80% / worst-20% split follows a stated rule. Disconnected outputs are shown (Fig. 13) and said to occur occasionally, but their number is not reported.

P00165 · Abueidda et al. (2020)

Topology optimization of 2D structures with nonlinearities using deep learning. arXiv preprint. Version of record: Computers & Structures 237 (2020) 106283, doi 10.1016/j.compstruc.2020.106283 (Crossref; same three authors and title; Elsevier user licence, not open access per OpenAlex; not accessed). DOI 10.1016/j.compstruc.2020.106283. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On held-out test sets, ResUnet predictions thresholded at 0.5 reproduce the TO ground-truth designs with mean Dice similarity 0.958 (linear elastic, 32×32); 0.964 (neo-Hookean) and 0.984 (stress-constrained), both 50×50. **Location:** Sec. 5, pp. 29-30 (linear model; first quantitative comparison with classical TO results); neo-Hookean pp. 32-33; stress-constrained p. 34. The abstract, introduction and conclusion compare with TO only qualitatively

1. **R1: partial** (R1-loose). Sec. 3 and 5: predictions and TO designs share the same inputs (load position, angle and magnitude, volume target, filter radius), so the comparison is at matched problem settings. Matching is loose: agreement is measured only by DSC on thresholded images; the volume fraction and compliance of the predicted designs are not reported; the TO stopping rules are not stated; no time or FE budget is matched.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (pp. 1-2) names no baseline beyond 'a topology optimization framework' and gives no figures, so (b) is not observed. For context: the linear and neo-Hookean reference designs come from ABAQUS/TOSCA, a general-purpose commercial TO module (Sec. 2.1, p. 9), and about 25 min per 32×32 linear optimization on a Core i5 PC (Sec. 3.1, p. 17; about 2,200 DOFs [derived]) looks slow for the size [inferred]; iteration counts are not stated. Both coders coded partial (R2-inefficient-impl) before the stratum decision.

3. **R3: na** (R3-na-quality-stratum). Primary outcome is quality only. For C3: the qualitative speed claim concerns inference time only (abstract). Training 1.25 h and 1.5 h on one V100 (Sec. 5, p. 26); data 15,000 and 18,000 samples at 0.31 and 3.2 min per data point on ten parallel iForge tasks (Sec. 3.1) [derived: 77.5 h and 960 h]; stress model: neither cost stated; no break-even. [derived] Because the data are TO runs, the linear model breaks even only after about 15,000 uses at PC rates (15,000 × 25 min of data vs 25 min saved per use), plus about 3 uses for training.

4. **R4: fail** (R4-image-metric). Sec. 4.2 and 5: design quality rests only on the Dice similarity of thresholded images (0.958, 0.964, 0.984) and on randomly picked visual examples (Figs. 10, 11, 13). No FE analysis of any predicted design is reported: no compliance, no volume check and no stress-constraint check for the stress model. Disconnections are said not to occur for the linear model (p. 32), without a count.

P00176 · Zhang et al. (2019)

A deep Convolutional Neural Network for topology optimization with strong generalization ability. arXiv preprint (cs.LG; also stat.ML). A retitled version with a partly different author list is Engineering Optimization 54(6) 973-988, doi 10.1080/0305215X.2021.1902998 (Crossref) [same work inferred from title, overlapping authors and publisher abstract; paywalled, not accessed]. DOI 10.48550/arXiv.1901.07761. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On 8000 held-out 40×80 cantilever problems, the U-Net predicts a layout in 0.001 s on average versus 3.913 s for SIMP (88-line code), with mean compliance error 4.16% after excluding the 4.12% of outputs that are disconnected. **Location:** Sec. 5, p. 7, and Table 1 (first quantitative classical comparison; the abstract, conclusion and introduction make only qualitative claims)

1. **R1: fail** (R1-unmatched). Sec. 5 p. 7 and Table 1: mean CNN inference time (0.001 s) is compared with mean SIMP time (3.913 s) on the same test problems, while the CNN designs are measurably worse than the SIMP designs they imitate: mean compliance error 4.16%, and 4.12% of outputs are

disconnected with huge compliance errors and are excluded from all averages. No matching is attempted (e.g., SIMP stopped at equal compliance); the SIMP stopping rule is not stated.

2. **R2: pass** (R2-none-established). Sec. 4.1 (p. 4) and Sec. 5 (p. 7): SIMP is the 88-line MATLAB code, and all calculations of both methods ran on one PC with an i7-8700K CPU and a GTX 1080 Ti GPU; the device for each method is not stated. The problem has about 6.6×10^3 DOFs [derived], below 10^5, so (c) does not apply. The abstract gives no baseline figure (b). No iteration count or stopping rule is given, so (d) and (e) cannot be derived. The 88-line code is a vectorized educational code [recalled, not verified in-session]; no inefficiency is evident. Unsure, so pass.

3. **R3: fail** (R3-no-cost). Sec. 5 p. 7: the authors state that the considerable dataset-generation and training times are not included and argue that they become negligible over many uses; neither is quantified and no break-even is given. Dataset: 80,000 SIMP samples (Sec. 4.1) [inferred ≈87 h at the 3.913 s mean], plus one FE solve per sample for the input tensor; training time not stated.

4. **R4: pass** (R4-FE-verified). Table 1 and Sec. 5: compliance, pixel and volume-fraction errors are computed for all 8000 test designs against SIMP (compliance by FE analysis [inferred]; error definitions not given). Volume-fraction error (0.13%) is reported, and disconnected outputs are counted (4.12%; 13.64% and 11.50% for the OOD sets, Table 2), although they are excluded from the averages. Figures show examples, but the numbers cover the full test sets.

P00178 · Rawat and Shen (2019)

A Novel Topology Optimization Approach using Conditional Deep Learning. arXiv preprint. DOI 10.48550/arXiv.1901.04859. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). After training, the CWGAN produces a 120×120 cantilever design in about 0.34–0.35 s versus about 71–115 s for MATLAB SIMP (e.g. 0.350 s vs 83.914 s), roughly two orders of magnitude faster, claimed without loss of optimality. **Location:** Results and Discussions, Table 1 and the paragraph after it, p. 6 (abstract, conclusion and introduction hold only qualitative claims)

1. **R1: fail** (R1-unmatched). Results p. 6, Table 1: the speed claim (0.350 s vs 83.914 s at target volume fraction 0.3) compares CWGAN inference with completed SIMP runs, while the authors report that generated designs stay near volume fraction 0.4 whatever the target, so the cited 0.3 output misses its constraint. No compliance of any CWGAN design is computed and no matching is attempted. Whether the post-processing (0.5 threshold plus Gaussian filter, p. 5) is inside the 0.350 s is not stated.

2. **R2: partial** (R2-hardware-unjustified; R2-inefficient-impl [inferred]). Dataset p. 5: SIMP runs in MATLAB, and the article cites the 99-line and 88-line educational codes (refs 27, 28) [inferred as the code base]. Hardware, linear solver and iteration counts are not stated for either method. The problem is 120×120 (below 10^5 DOFs), so (c) does not apply; (b) and (d) are not established by the text.

3. **R3: fail** (R3-no-cost). The speed claim (p. 6) reports neither training time nor data-generation cost. Only the dataset size is given (3024 SIMP samples, Dataset p. 5); at the Table 1 SIMP times this is about 60–96 h of generation [inferred: 3024 × 71.461–114.635 s]. No break-even and no per-query limitation.

4. **R4: fail** (R4-selected-examples). Table 1 and Figs. 5–6: quality is judged visually on one generated design per target, chosen without a stated rule. No FE analysis of any generated design, no compliance and no measured volume fraction. The conclusion (p. 7) notes that GANs lack a proper evaluation metric.

P00181 · Banga et al. (2018)

3D Topology Optimization using Convolutional Neural Networks. arXiv preprint (arXiv comment: under review for a Computer-Aided Design special issue on generative design; no journal reference on the abs page). DOI 10.48550/arXiv.1808.07440. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Stopping 3D SIMP (TopOpt) at iteration 20 and letting a 3D CNN predict the final design cut computation time by about 40% versus solving to convergence, with 96.2% binary and 79.7% RMS voxel accuracy on 1,000 test problems. **Location:** Abstract (quantitative comparison with the conventional solver alone); restated in Sec. 5.4 (p. 13, 40-45%) and the Conclusion (pp. 13-14, 40%)

1. **R1: partial** (R1-loose). Abstract (p. 1), Sec. 5.4 (p. 13), Conclusion (pp. 13-14): the roughly 40% time saving is measured against running TopOpt to convergence, while the CNN outputs are compared with the converged designs only by voxel accuracy (96.2% binary, 79.7% RMS; Sec. 5.1, p. 12). No compliance or volume of any predicted design is computed, and SIMP is not stopped at matched accuracy; per Fig. 6 (p. 9), raw SIMP iterates already exceed 90% binary accuracy early, so the CNN's gain over early stopping is untested. The TopOpt stopping rule is not stated (70-100 iterations to convergence, Sec. 4.1, p. 6). The authors call the network output sub-optimal and list island removal as future work (Sec. 6, p. 14), but no design-level gap is measured, so the fail marker (measurably worse output) is not established. A stricter reading (speed claim with no design-quality comparison) gives fail.

2. **R2: pass** (R2-none-established). Sec. 4-4.1: the baseline is the public PETSc TopOpt code (ref. [31]), run to convergence (70-100 iterations); its default solver is multigrid-preconditioned FGMRES, the protocol's efficient class [observed in the public code; settings used here not stated]. Hardware is not stated. The problem has about 1.3×10^4 DOFs [derived], so (c) does not apply; (b) not observed (the abstract and results use the same comparator); (d) not established (convergence-based stopping); (e) not derivable (no times). Unsure, so pass.

3. **R3: fail** (R3-no-cost). Sec. 1 notes qualitatively that the approach trades a large offline data corpus for run-time speed, but neither the cost of the 6,000 TopOpt runs (Sec. 4.1) nor the training time (App. A: 30 epochs) is reported, and no break-even is given. For C3: with about 6,000 full solves of data and a saving of about 40% of one solve per use, break-even is about 15,000 uses before training [derived: 6,000 ÷ 0.4].

4. **R4: fail** (R4-image-metric). Sec. 5 and Tables 3-7: quality rests only on voxel binary and RMS accuracy against the solver output. No FE analysis of any predicted design, no volume check and no count of disconnected designs are reported; the conclusion lists avoiding islands and controlling volume fraction as future work (Sec. 6). Table 7 and App. C show selected examples.

P00183 · Cang et al. (2019)

One-Shot Generation of Near-Optimal Topology through Theory-Driven Machine Learning. Computer-Aided Design 109 (2019) 12–21. DOI 10.1016/j.cad.2018.12.008. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The trained one-shot generator produces a topology in the order of 10^{-2} s per problem on a Xeon E5-1620 CPU, versus 10^2 to 10^3 s per problem for the TO solver; the authors exclude data-acquisition cost from this comparison. **Location:** Section 5.4, paragraph on solution generation speed, p. 13 (abstract and conclusion hold only qualitative ML-vs-classical claims; their quantitative claims are ML-vs-ML)

1. **R1: fail** (R1-unmatched). Section 5.4 p. 13: generator time (about 10^{-2} s) is compared with full TO solves (10^2–10^3 s) run to the AL tolerances (Alg. 2). The generated designs are measurably worse: Case 2 median compliance gap 5.2 with std 0.6 over 10 runs and failure rate 0.16% (Table 1); Case 1 mean gap 6.2 (Fig. 3b), 3.5–16.7% on the Fig. 4 examples [derived]. Mean global volume violation is 8.76% vs 4.89% for ground truth in Case 1 (1.6% vs 3.0% in Case 2) (p. 12). No classical run is matched to the generator's quality or time.

2. **R2: partial** (R2-inefficient-impl). Section 3.2 p. 6: the AL solver needs about 5000 FE analyses on average for a 4800-element 2D problem; Alg. 2 (Appendix, p. 17) uses gradient steps with a 10^{-3} learning rate, backtracking, step clipping and β-continuation, and the Appendix (p. 16) notes that MMA is also applicable. The resulting 10^2–10^3 s per 2D topology (Section 5.4, p. 13) makes the classical implementation look inefficient. Typical OC/MMA runs on such problems need far fewer FE solves [recalled, not verified in-session]. Linear solver and TO-solver hardware are not stated. (a) and (e) need a re-test (per-FE-analysis time ≈0.02–0.2 s is derivable [derived: 10^2–10^3 s ÷ 5000]); (b)–(d) are not established.

3. **R3: partial** (R3-no-training-cost). Section 5.4 p. 13: the authors state that generator run time excludes data acquisition and that the comparison holds only when the expected TO workload far exceeds the training cost (an explicit per-query limitation). Data cost is given as an FE-analysis budget (Case 2 equals 7000 TO solves, p. 11), but NN training time is not reported and no break-even is computed; Section 6.4 discusses the trade-off only qualitatively.

4. **R4: pass** (R4-FE-verified). Eq. (8), Figs. 3–6, Table 1: compliance of every generated test design is computed by FE analysis and compared with the AL ground truth over 100 (Case 1) and 1000 (Case 2) test inputs, 10 runs each. Global and local volume-constraint violations are reported (p. 12), and failed designs (gap $>$ 1000) are counted for Case 2 (Table 1). Case 1 failure counts and the number of volume-violating designs are not given.

P00193 · Kwon and Kang (2026)

Zero-shot rib design: merging training-free generative prior with topology optimization. arXiv preprint (cs.LG; also physics.app-ph), CC BY-NC-ND 4.0; page footer 'Preprint submitted to Elsevier'. Version of record: Advanced Engineering Informatics 77, article 105276, doi 10.1016/j.aei.2026.105276, retitled 'Zero-shot rib design: Merging a training-free generative prior with topology optimization' (Crossref); pool record P01925. Crossref lists only Elsevier TDM and STM policy licences, and OpenAlex marks the work open only through arXiv, so the version of record is not open access and was not read. DOI 10.48550/arXiv.2609.10643. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Across 245 SDS runs (four domains; mechanical and thermoelastic compliance), 38 of 49 prompt-domain combinations beat the single Adam-SIMP baseline on all five seeds, with reductions up to 31.5% (mechanical, best run) and 23.0% (thermoelastic, five-seed mean). **Location:** Abstract (first quantitative comparison with a classical method, §5.1 rule 1); numbers in Sec. 5.1 and Table 5 (mechanical) and Sec. 5.5 and Table 7 (thermoelastic)

1. **R1: partial** (R1-loose). Stratum Q; R1 applies as written (PROTOCOL §11, 2026-09-24). Sec. 3-5, App. B.3, App. D.1: each SDS run and the Adam-SIMP baseline share mesh, filters, projection schedule, exact volume and 1,000 iterations, and per-run wall-clock is close (about 44 vs 39-41 min [derived]). The headline is loosely matched: significance compares five SDS seeds with one deterministic baseline run; 'up to -31.5%' is the best of 50 runs per domain, while the best of 50 run-matched multi-start SIMP runs (47.434, Table D.1, p. 32) leaves a D_Hole gain of -8.9% [derived]; and prompts, the SDS weight per domain and the schedule were chosen on pilot runs (App. D.1, D.6), a tuning budget the baseline does not get. App. D.1 matches run counts only for the multi-start and perturbation baselines.

2. **R2: fail** (R2(b)). Stratum Q; R2 applies through criterion (b) (PROTOCOL §11, 2026-09-24). Abstract (p. 1) vs Sec. 5.2 and Table 6 (p. 13): every figure in the abstract (38 of 49 significant; up to -31.5% and -23.0%) is measured against the Adam-SIMP baseline (Tables 5 and 7), and the abstract says SDS outperforms gradient-based baselines. Table 6 reports a tuned MMA baseline, which the text calls a standard strong baseline; its compliance is 15-23% below Adam's [derived], and it beats the best SDS run in D_Rect. Against MMA the best D_Hole gain falls from -31.5% to -11.6%, and no prompt mean beats MMA in D_Rect or D_Circle [derived from Tables 5-6]. The authors note that MMA used about twice the compute per iteration (4.90 vs 2.34 s, p. 13).

3. **R3: na** (R3-na-quality-stratum). The primary outcome is quality only, and the method is per-problem optimization with a frozen pretrained prior, so there is no offline data or training cost (Sec. 1, Sec. 3, Data availability). For C3: each SDS run takes about 7% more wall-clock than the Adam-SIMP baseline (Sec. 4).

4. **R4: pass** (R4-verified). Sec. 3.1, Sec. 6.4, Tables 5 and 7, App. B.3, App. D.1-D.4, App. D.8: compliance is the FE compliance of the returned physical density field (gray ratio below 3%) at exactly the target volume. Every seed is shown (Figs. D.2-D.4), and non-significant and degrading prompts and the heat-conduction negative result are reported. The CAD post-processing is not used for the reported compliance; the 3D solid re-analysis checks rankings only (r = 0.82-0.98). The MMA compliance in Fig. 5 is the best value over the run, not the final value.

P00309 · Xu et al. (2024)

Evolutionary de-homogenization using a generative model for optimizing solid-porous infill structures considering the stress concentration issue. arXiv preprint (math.OC); version of record 2025 at doi 10.1016/j.matdes.2025.114380 (Materials & Design [inferred from the DOI prefix]; not accessible, HTTP 403). DOI 10.1016/j.matdes.2025.114380. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On the symmetric tension beam, the VAE-driven evolutionary stage (MFTD) cut maximum von Mises stress by 13%, 15.6% and 18.2% versus classical low-fidelity stress-TO

designs at similar mass ratios; hypervolume rose 35.5% over 200 iterations. **Location:** Section 4.2.3, Figs. 20-21 (no quantitative classical comparison in abstract, conclusion or introduction)

1. **R1: partial** (R1-loose). Sec. 4.2.3 (p. 29), Fig. 21: stress comparisons at approximately matched mass ratio (10.27 vs 10.33%, 12.41 vs 12.3%, 17.1 vs 16.7%); by the printed values MFTD uses more material in two of the three pairs, although the text reports weight reductions [inferred]. No computational budget is reported: MFTD ran 200 iterations with high-fidelity FEA of candidates, and the stopping rule of the classical low-fidelity runs is not stated (Sec. 2.1.4; Table 1, p. 17).

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no baseline and gives no figures, so (b) is not observed. No runtimes are reported for any method, so criteria (a) and (c)-(e) could not be judged in any case.

3. **R3: na** (R3-na-per-problem-learning). Stratum Q (PROTOCOL §11, 2026-09-24): R3 is na. No speed or cost claim is made. The VAE is retrained at every iteration inside one design run, with no offline training (Sec. 3.2.4), so its cost sits inside R1/R2.

4. **R4: partial** (R4-no-failure-rate). Quality = maximum von Mises stress from COMSOL adaptive-mesh FEA of the de-homogenized geometry, with a mesh-independence study (Sec. 2.2.3, 4.1.2); mass ratio reported for each design. Anomalous candidates are discarded without a reported count (Sec. 3.2.2), and the three per-design pairs use chosen mass ratios without a stated selection rule; the hypervolume comparison covers all Rank-1 designs.

P00358 · Krishnamurthy (2024)

Latent Diffusion Models for Structural Component Design. Computer-Aided Design 171, 103707 (2024), per Crossref search; arXiv preprint coded. DOI 10.1016/j.cad.2024.103707. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). For 100 test conditions (2,000 generated 3D designs), the distribution of surrogate-predicted strain energy shows no marked aggregate difference from the 100 matching SIMP designs. On this basis the paper presents the generated designs as near-optimal. **Location:** Sec. 6.2 Strain Energy Analysis, Fig. 8 (first quantitative performance comparison with SIMP); the Abstract and Conclusions give only the qualitative near-optimality claim

1. **R1: partial** (R1-loose). Stratum Q, so R1 applies as written (PROTOCOL §11, 2026-09-24). Each set of 20 generated designs shares the initial strain energy (load case) of its SIMP design (Sec. 6, p. 6). Volume fraction is not matched: the only LDM input is the initial strain-energy grid (Sec. 4, p. 4), the Conclusions name volume-fraction conditioning as future work (Sec. 7, p. 9), and the average generated volume fraction differs from the SIMP design by a mean absolute error of 0.0989 (Sec. 6.2, Fig. 7, p. 7) for targets of 10-50 % (Sec. 5, p. 6). The SIMP stopping rule is not stated (Sec. 5). The strain-energy comparison is of aggregate, surrogate-predicted distributions (Fig. 8; see R4). Quality is reported but matching is loose: the §5.3 partial case. No speed claim is made.

2. **R2: na** (R2-na-no-speed-claim). The primary outcome is quality only. SIMP (ANSYS Mechanical APDL v19.2, Sec. 5) supplies the reference designs, and its run time is never compared.

3. **R3: na** (R3-na-no-speed-claim). There is no speed claim. For reference, training times are reported per epoch or per sample (Sec. 6.4); SIMP data-generation time is not stated; voxelization takes about 15 minutes per geometry (Sec. 5).

4. **R4: fail** (R4-surrogate). Sec. 6.2, Fig. 8: strain energy of the generated designs is predicted by a surrogate CNN whose accuracy is not reported. No FE analysis of any generated design is reported. Generated volume fractions deviate from the SIMP designs (MAE 0.0989). Floating or disconnected designs are not counted; floating artifacts are mentioned only for a rejected encoder variant (Sec. 4.1).

P00373 · Mallon et al. (2023)

Neural Level Set Topology Optimization Using Unfitted Finite Elements. arXiv preprint (cs.CE). Version of record: International Journal for Numerical Methods in Engineering (2025), doi 10.1002/nme.70004 (hybrid open access, CC BY-NC-ND per OpenAlex; not accessed, see notes). DOI 10.1002/nme.70004. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Fig. 7: with the same mesh, volume fraction and unfitted evaluator, the NN-LS design had the lowest heat-conduction objective (pixel SIMP 15.7% and pixel LS 20.3% worse) but was 2.6% worse than pixel SIMP on the MBB beam. **Location:** Sec. 7.1.2, Fig. 7 (p. 15): the first quantitative comparison with classical methods. The abstract, introduction and conclusion compare only qualitatively

1. **R1: partial** (R1-loose). Stratum Q, so R1 applies as written (PROTOCOL §11, 2026-09-24). Sec. 3 (p. 7), Sec. 7.1, Figs. 7-8: same meshes, volume fraction 0.4 and a common unfitted evaluator. Budget is not matched: NN-LS reports the best of 100 random seeds, while seeding for the pixel baselines is not stated [inferred: single runs]; the optimizers differ (ADAM vs MMA); stopping rules and times are not stated for any method (Fig. 8 shows 0-1,000 function calls). The volume of the SIMP designs after 0.5 iso-surface conversion is not reported (Sec. 7.1.2, p. 14). The bias correction of Sec. 7.1.3 applies only to the Fig. 8 convergence curves. The heat-conduction gain may partly reflect the best-of-100 selection [judgment].

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The arXiv v2 abstract (p. 1) names no baseline and gives no baseline figures; the version-of-record abstract in the pool record mentions only unnamed baseline approaches with similar performance. (b) is not observed. For context: pixel SIMP (penalization 3) and pixel LS use MMA on the same meshes (Sec. 7.1, p. 13), and no run time is reported for any method.

3. **R3: na** (R3-na-per-problem). Per-problem neural reparameterization with no dataset and no offline training (Sec. 2-3); no speed claim. For C3: no runtimes are reported; the method needs a similar or larger number of function calls than SIMP (Fig. 8), and 100 random seeds per problem (Sec. 3, p. 7).

4. **R4: partial** (R4-no-constraint-report). Sec. 7.1.2 and Fig. 7: every design, including SIMP after 0.5 iso-surface conversion, is re-evaluated with the same unfitted FE solver. The LS volume is enforced exactly by the translation step (Sec. 4.4), but the volume of the converted SIMP designs is not reported, and the spread or failures over the 100 NN seeds are not reported; only the best design is shown (selection rule stated: lowest objective).

P00375 · Giannone and Ahmed (2023)

Diffusing the Optimal Topology: A Generative Optimization Approach. arXiv preprint (cs.LG; cs.CE). Also published as an ASME IDETC-CIE 2023 conference paper (49th Design Automation Conference, Vol. 3A, V03AT03A012, 20 Aug 2023; Crossref record of doi 10.1115/detc2023-116595, ASME standard licence; not read, see notes). DOI 10.1115/detc2023-116595. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On in-distribution test constraints (task-1, 64×64), diffusion designs conditioned on the cheap kernel and refined by 10 SIMP iterations have 4.16% mean compliance error relative to the SIMP-optimized references, with 1.16% volume-fraction error and 5.61% floating material. **Location:** Sec. 5.1, Table 5, p. 9 (first quantitative comparison with a classical method; the Abstract, Sec. 7 Conclusion and Introduction hold no quantitative classical comparison, and the Introduction's 53.93%, 23.81% and 25.64% figures are against ML baselines)

1. **R1: partial** (R1-loose). Stratum Q, so R1 applies as written (PROTOCOL §11, 2026-09-24). Table 5 and Sec. 5 Evaluation: generated designs are compared with SIMP references at the same prescribed volume fraction, but volume is only approximately matched (VFE 1.16%). Samples with compliance $>$ 100 are removed before averaging, with no count reported. Budgets are not matched or stated: the ML side spends 2.55 s of sampling, processing and 10 SIMP iterations; the SIMP reference runs have no stated iteration count or time. No time is compared with SIMP anywhere (Tables 5-7 time ML models only; Table 9 reports no time), so the R1 fail markers, which need a reported time, do not apply.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names SIMP only as computationally expensive and prone to local minima, and gives no baseline figures, so (b) is not observed. For context: the SIMP references come from the TopoDiff dataset [13], and this article states none of their settings (iterations, continuation, solver); Table 9 stops SIMP at 5 or 10 iterations from an unstated initial design.

3. **R3: na** (R3-na-no-speed-claim). No speed or cost claim against a classical method. For C3: training is 200k steps on 30,000 SIMP-optimized topologies (Sec. 5 Setup); training time, data-generation cost and hardware are not reported.

4. **R4: partial** (R4-filtered-outliers). Compliance of generated designs is compared with SIMP compliance for the test configurations, presumably by FE analysis [inferred; the evaluation procedure is not described]. VFE, floating-material rate and load disrespect are reported (Tables 5-7). But Tables 5-6, including the primary outcome, drop samples with compliance > 100 and do not report how many; only task-3 (Table 7) is unfiltered. Figs. 11-12 show selected bad and good examples without a stated selection rule.

P00527 · Jeong et al. (2023)

A Physics-Informed Neural Network-based Topology Optimization (PINNTO) framework for structural optimization. Engineering Structures 278 (2023) 115484 (online 2022-12-21 per OpenAlex). DOI 10.1016/j.engstruct.2022.115484. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On four 40×20 compliance benchmarks, PINNTO designs reach compliance within 0.02-5.51% of 88-line SIMP designs (lower only in case 1), while each run takes 399-2,177 s against 0.3-1.6 s for SIMP. **Location:** Sec. 4.1, Fig. 7 and Table 1 (first quantitative comparison with a classical method; the Abstract claims comparable designs to SIMP without numbers, and Sec. 5 has no numbers); key phrase from the Abstract

1. **R1: pass** (R1-matched). Sec. 4.1, Fig. 7, Table 1: both methods use the same 40×20 meshes, volume fractions, rmin = 3, p = 3, OC update and convergence tolerance; iteration counts and times are reported for both. Compliance is within 0.02-5.51%, PINNTO lower only in case 1. The parity claim is made at about 1,100-1,400 times the SIMP wall-clock [derived], so the unequal budget favours the ML side and is fully disclosed. [judgment: an adjudicator may prefer partial for the unequal budget]

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (AAM p. 1) names SIMP as the comparator and gives no numbers; the results use the same SIMP family (88-line code in 2D, Liu and Tovar code in 3D; Sec. 4.1, 4.6), so the abstract does not report a weaker baseline than the results and (b) is not observed. For context: the SIMP baseline (88-line MATLAB, same CPU) is by far the faster method in Table 1.

3. **R3: na** (R3-na-per-problem). Secs. 3.1-3.2: the PINN is trained from scratch inside each TO iteration without labelled data; its cost is part of the Table 1 times (R1/R2). Also quality stratum.

4. **R4: partial** (R4-surrogate-predicted). Secs. 3.1.2, 4.1 and 4.3: PINNTO's compliance appears to come from the PINN displacement field [inferred]; no FE re-analysis of the returned PINNTO designs is described, and Sec. 4.3 explains PINNTO's lower coarse-mesh compliance by FEA inaccuracy, which suggests the two objectives come from different solvers. Achieved volume fractions and failure counts are not reported; Fig. 12 shows non-optimized results for small networks (qualitative only).

P00533 · Jeong et al. (2023)

A complete Physics-Informed Neural Network-based framework for structural topology optimization. Computer Methods in Applied Mechanics and Engineering 417 (2023) 116401 (issue dated December 2023, online 2023-09-09; CC BY-NC 4.0 per Crossref). DOI 10.1016/j.cma.2023.116401. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract states that CPINNTO, a deep-energy-method PINN for analysis plus a sensitivity PINN, reaches optimal topologies without labelled data or FEA, with compliance comparable to density-based TO designs (no numbers). **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no marker). no full text

2. **R2: na** (no marker). no full text

3. **R3: na** (no marker). no full text

4. **R4: na** (no marker). no full text

P00537 · Li et al. (2019)

Non-iterative structural topology optimization using deep learning. Computer-Aided Design 115 (2019) 172-180 (online 2019-05-29; closed access per OpenAlex). DOI 10.1016/j.cad.2019.05.038. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract claims that a two-stage predictor, a GAN for a low-resolution layout plus an SRGAN for refinement, trained on many TO runs, estimates conductive heat-transfer topologies accurately in negligible computational time; no numbers are given. **Location:** Abstract (LSU repository record page)

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P00549 · Kudela et al. (2023)

Deep learning aided topology optimization of phononic crystals. Mechanical Systems and Signal Processing 200 (2023) 110636 (available online 30 July 2023; CC BY 4.0). DOI 10.1016/j.ymssp.2023.110636. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). A CNN surrogate predicts a phononic unit cell's full dispersion diagram in under 100 ms, versus roughly 0.5-1 h for an FEM (COMSOL) calculation, so a genetic algorithm can use it inside the topology optimization loop. **Location:** Sec. 4 Conclusions, pp. 12-13 (first quantitative classical comparison; the abstract and introduction are qualitative); supporting timing example in Sec. 2.1, p. 4 (about 47 min for one 134,040-DOF FEM run)

1. **R1: partial** (R1-loose). Sec. 2.3, 2.6 and 4: surrogate accuracy is reported against FEM on all 300 test samples (Table 2), so quality is reported at the level of the claim. Matching is loose. The quoted 0.5-1 h (and the 47 min example) concern FEM runs that appear finer or slower than the simplified-geometry FEM that produced the training labels: the dataset works out to ≈288 s per sample [derived]. The article itself says the cavity geometry was simplified to cut COMSOL time. No optimization-level comparison exists.
2. **R2: partial** (R2-inefficient-impl). Sec. 2.1, 2.3: the classical cost comes from general-purpose commercial FE (COMSOL via MATLAB LiveLink) on an 8-core CPU. The quoted per-evaluation time is 6-12× the ≈288 s implied by the authors' own dataset pipeline [derived], and the surrogate's hardware is not stated. (c) cannot be established: the example has 134,040 DOFs, but whether the surrogate ran on a GPU is not stated. (b) not observed: the abstract has no figure. (d) and (e) not applicable.
3. **R3: partial** (R3-no-training-cost). Sec. 2.3: data generation is reported (about one month for 9,000 samples); training time is not, and no break-even is given. Sec. 1 notes qualitatively that the speed comes at the cost of supervised training. C3 [derived]: at ≈288 s per FEM sample, the data cost equals 9,000 FEM evaluations; at the quoted 0.5-1 h, 720-1,440 evaluations. One GA run allows up to 20,000 evaluations (400 × 50), so the data cost is repaid within about one run, training excluded.
4. **R4: partial** (R4-no-failure-rate). Table 2: surrogate outputs are FE-checked on every test sample. Optimized cells are re-analysed with FEM, but only as curve overlays (Figs. 8-11); the Table 3 values are surrogate predictions. Wall-thickness feasibility is discussed (Sec. 3.2, Fig. 12); one surrogate misprediction is disclosed (Fig. 9) without a rate. A stricter reading gives fail: the two runs shown per objective were picked from several GA runs, with no stated selection rule (Sec. 3.2).

P00550 · Senhora et al. (2022)

Machine learning for topology optimization: Physics-based learning through an independent training strategy. Computer Methods in Applied Mechanics and Engineering, Vol. 398, Art. 115116. DOI 10.1016/j.cma.2022.115116. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The abstract claims that the offline-trained CNN sensitivity surrogate makes topology optimization up to 30 times faster than traditional topology optimization without loss of

accuracy. The largest measured speed-up is 26.5x, on a 10.6-million-element MBB beam with block size 3 (39 h vs 1.5 h, Fig. 16). **Location:** Abstract (p. 1; first quantitative comparison with a classical method, §5.1 rule 1). The measurements behind it are in Sec. 6.4 and Fig. 16 (pp. 18-20); the Introduction states about 27x (p. 4) and the Conclusion up to 26 times (p. 19)

1. **R1: partial** (R1-loose). Both methods use the same 200-iteration continuation schedule and volume fraction 0.12 (Sec. 6, p. 14; Sec. 6.2). At 1.3 million elements the timed runs can be matched to Fig. 13 (p. 18): ML 15.6 (NB = 2) and 15.7 (NB = 3) against reference 15.4, i.e. +1.3% and +1.9% [derived]. The headline 26.5x case (10.6 million elements, NB = 3; Fig. 16, p. 20) shows only the ML compliance (10.9, Fig. 12, p. 17), with no reference value, so the largest speed-up is not tied to a matched-quality result. Quality is reported, but matching is loose (§5.3 partial).
2. **R2: pass** (R2-gpu-baseline). Sec. 6.4: the standard/traditional TO baseline and the ML framework run on the same machine (24 Intel Xeon CPUs, 251 GB RAM, 2 NVIDIA Titan Xp GPUs), and the baseline's fine-mesh FEA itself uses the group's in-house CUDA (GPU-accelerated) solver, not a CPU-only or naively-vectorized implementation, so there is no GPU-vs-CPU-only asymmetry. The same iteration cap and continuation schedule apply to both methods. Solver type (direct/iterative/multigrid) is not named, so full confidence on solver efficiency is not possible, but nothing in the article indicates an inflated iteration count or an artificially weakened baseline. For a later R2(e) check, per-iteration times are derivable from Fig. 16: about 50 s at 1.3 million elements and 11.7 min at 10.6 million elements [derived, assuming 200 iterations].
3. **R3: fail** (R3-no-cost-reported). No wall-clock time, GPU-hours, or any other cost figure is given anywhere in the article for either training the CNN models or generating the training data (Sec. 6.1 describes data generation only qualitatively as 'relatively more efficient'; Appendix B shows only loss-vs-epoch curves). The 26.5x/30x speed claims are made with no quantified upstream cost and no break-even analysis at all.
4. **R4: partial** (R4-partial-verification). Compliance is computed by FE analysis of the returned design for the moderate-size examples (Sec. 6.2, Fig. 13, 'reference compliance' compared directly), and the article explicitly states the framework avoids 'hanging members nor disconnected components' in its examples (Sec. 6.2), a qualitative feasibility check. However, for the largest (38-million-element) example, fine-mesh compliance could not be computed and only a coarse-mesh estimate is reported (Sec. 6.3, self-disclosed), and no quantified failure or infeasibility rate is given for any of the examples.

P00597 · Sun et al. (2022)

Topological dimensionality reduction-based machine learning for efficient gradient-free 3D topology optimization. Materials & Design 220 (August 2022) 110885 (Crossref). Gold open access: Crossref lists a CC BY-NC-ND 4.0 licence for the version of record. DOI 10.1016/j.matdes.2022.110885. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract claims that a material-field series expansion (under 200 design variables) with a sequential DNN surrogate cuts computing time by 1-4 orders of magnitude versus coarse-mesh gradient-free methods, enabling gradient-free 3D TO with thousands of elements. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P00601 · Gavris and Sun (2024)

Topology optimization with graph neural network enabled regularized thresholding. Extreme Mechanics Letters 71 (2024) 102215 (online 2024-08-23; bronze OA per OpenAlex; Crossref lists an Elsevier open-access licence for the accepted manuscript from 2025-08-23). DOI 10.1016/j.eml.2024.102215. Stratum: none; access: no access; stage: agreed.

Primary outcome (paraphrase). Not coded: no accessible full text or abstract. **Location:** not available

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P00605 · Liu et al. (2023)

Fast topology optimization of phononic crystal-based metastructures for vibration isolation by deep learning. Computer-Aided Civil and Infrastructure Engineering 39(5) (2024) 776-790 (Crossref; published online 2023-09-30 per the pool record; hybrid open access, CC BY-NC-ND 4.0 per Crossref and OpenAlex). The DOI now resolves to Elsevier (pii S1093968726016749). DOI 10.1111/mice.13100. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: the deep-learning-based optimization (DLBO) reaches bandgap results similar to or better than a genetic algorithm while needing 0.01% of the GA's time. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P00610 · Adam et al. (2023)

Effective thermal conductivity estimation using a convolutional neural network and its application in topology optimization. Energy and AI, vol. 15, article 100310 (the article's own running header prints 'Energy and AI 15 (2024) 100310'; first page states 'Available online 19 October 2023', received 25 Jul 2023, accepted 14 Oct 2023; open access under CC BY-NC-ND). DOI 10.1016/j.egyai.2023.100310. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The CNN predicts the effective thermal conductivity of a 128x128 two-phase image in 15 ms, which the abstract calls 3 to 5 orders of magnitude faster than the finite-volume simulation that each iteration of the voxel-swap topology optimization would otherwise need. **Location:** Abstract (first quantitative comparison with a classical method); the same comparison is elaborated with exact numbers in Sec. 4.2 (CPU-vs-CPU, Table 1) and Sec. 5.1-5.4 (GPU-based per-iteration speed-ups inside the topology-optimization case studies)

1. **R1: pass** (R1-matched-per-evaluation). The primary outcome is a per-evaluation comparison (abstract; Sec. 4.2, p. 7), and at that level the CNN is checked against the same NHT solver it replaces: test MAPE 0.35-2.35% on 26,000 held-out images per conductivity ratio (Table 5, Fig. 5, p. 7), plus the analytical parallel and series cases (Sec. 4.3). The comparison therefore sits within the ML method's reported error of the classical result, as in adjudicated record P00149. In the optimization application the CNN- and NHT-driven variants share the voxel-swap and simulated-annealing optimizer and the budget of 1,000,000 attempts (Sec. 5, p. 7), and every accepted CNN swap is re-evaluated by NHT (Figs. 7-10). The NHT-driven run itself was only extrapolated (102,778 h), not executed.
2. **R2: partial** (R2-inefficient-implementation). The state equation (2D steady-state heat conduction) is elliptic, matching R2's scope. The classical NHT solver uses TDMA line-sweeps applied iteratively (not a sparse direct solve of the full system nor a multigrid-preconditioned Krylov method), and Table 1 reports 32,903-154,307 iterations to converge a single 16,384-cell (128x128) image – a very large iteration count for a linear 2D Poisson-type problem of this size, consistent with the partial condition 'the implementation looks inefficient... but no direct comparison is possible' (no re-test is performed in this coding pass, so R2(a)/(e) cannot be checked directly). Problem size (16,384 DOFs) is below the 10^5-DOF threshold named in R2(c), so the GPU (CNN, RTX 3070) vs. CPU (NHT, Bridges2 cores) asymmetry does not trigger an automatic fail, but it is also not justified in the article, and the CPU-vs-CPU claim in Sec. 4.2 ('using one CPU core, the CNN is 3 to 5 orders of magnitude faster') reuses Table 1's NHT times without confirming they came from the same CPU model (Ryzen 7 3700X)

used for the CNN's own CPU benchmark. Re-test pointer for (a)/(e): a sparse direct or multigrid solve of the same 128x128 finite-volume system on one core, compared with the 34.2-371.2 s per image of Table 1.

3. **R3: pass** (R3-full-accounting). Data-generation cost is reported concretely: 520,000 NHT simulations on a named supercomputer resource (Bridges2; Sec. 1, 2.3), with per-image times in Table 1 from which a total core-time is directly derivable. Training cost is reported concretely: about 1 hour per model on one HPC node with 4 NVIDIA V100 GPUs for 600 epochs (Sec. 4.2). The Conclusion (Sec. 6) explicitly states the cost relationship between them ('training of the CNN itself is minuscule... compared to creating a large dataset'), and every headline speed figure in Sec. 4.2 and Sec. 5 is explicitly scoped to per-image / per-iteration / per-batch inference, i.e. the claim is explicitly limited to per-query cost with both upstream costs disclosed rather than hidden.

4. **R4: pass** (R4-FE-verified). The returned designs of all four optimization runs, and every accepted swap on the way, are re-evaluated with the NHT finite-volume model and compared with the closed-form bounds (Sec. 5.1-5.4, Figs. 7-10, pp. 8-9). The porosity of 0.5 is kept exactly by the one-for-one swap rule (Sec. 5, p. 7), and all four runs are reported, none selected. Solid-phase connectivity is not a constraint of this two-phase conduction problem: Sec. 5.4 treats the checkerboard as the admissible maximum-surface-area state [judgment]. The CNN predictions are also shown against NHT for all 26,000 test images (Fig. 5), not only as averages. No unreported constraint violation or failure count is evident; unsure means pass.

P00624 · Rade et al. (2023)

Deep learning-based 3D multigrid topology optimization of manufacturable designs. Engineering Applications of Artificial Intelligence 126 (2023) 107033 (Elsevier; issue dated November 2023; Crossref lists an Elsevier open-access user licence for the accepted manuscript from 2024-09-07; OpenAlex status bronze). DOI 10.1016/j.engappai.2023.107033. Stratum: none; access: no access; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text and no abstract read). **Location:** none

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P00647 · Viswanath et al. (2024)

Designing a TPMS metamaterial via deep learning and topology optimization. Frontiers in Mechanical Engineering 10:1417606 (Original Research; open access, CC BY). Earlier preprint of the same study: arXiv 2303.10007 v1 (17 Mar 2023), 'Gyroid-like metamaterials: Topology optimization and Deep Learning', same six authors (pool P00826; arXiv API metadata read, full text not read). DOI 10.3389/fmech.2024.1417606. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). Once trained, the 3D ResUNet predicts an optimized 32^3 gyroid unit cell in 0.001 CPU-hours, versus about 2.4 h per homogenization-based SIMP run on one HPC core; accuracy is judged only by voxel MSE and Dice. **Location:** Sec. 3.2 (p. 8) and Table 2 (p. 10): the first quantitative ML-vs-classical comparison. The abstract and conclusion compare only qualitatively; the introduction (p. 2) gives the classical cost alone (about 67 s per iteration at 32^3, 200-800 iterations)

1. **R1: partial** (R1-loose). Sec. 3.2, Table 2, Sec. 5: DL prediction (0.001 CPU-h) is set against full TO runs (about 2.4 h each), and its outputs are compared with the TO ground truth only by voxel MSE (0.0079), Dice (about 95%) and volume deviation (0.24% mean). The bulk or shear modulus of a predicted cell is never computed, so objective-level matching cannot be checked; the authors note poorly learned discontinuous designs at low filter radius (p. 11). A stricter reading (speed claim with no design-quality comparison) gives fail (R1-no-quality).

2. **R2: partial** (R2-hardware-unjustified; R2(e) re-test pointer). Sec. 3.2 and Table 2: the classical time comes from single-core runs of a modified educational MATLAB code (chosen for parallel data

generation), while the prediction hardware is not stated. (c) does not apply: the timed prediction is on CPU and the problem has about 9.8×10^4 DOFs [derived]. (b) not observed (no figure in the abstract). (d) not established (convergence-based stopping, 1,000-iteration cap). For (e), a per-iteration time is stated: about 67 s at 32^3 with 6 load cases on a workstation (p. 2); it needs a same-size strong-baseline measurement. Unsure between pass and partial [judgment].

3. **R3: pass** (R3-costs-reported; per-query framing). Table 2 (p. 10) and Sec. 3.2 (p. 8) report data generation (24 h on 226 cores; about 2.4 h per run) and training (5.5 GPU-h) next to the prediction time. The abstract names the many costly TO runs as the model's major disadvantage and frames the gain as reuse after training. No break-even is given [derived for C3: about 2,750 uses = 6,600 core-h ÷ 2.4 h, before training]. A stricter reading (costs with no amortization) gives partial (R3-no-amortization).

4. **R4: fail** (R4-image-metric). Sec. 3.3 and Sec. 5, Figs. 13-16: quality rests on voxel MSE and Dice against the TO ground truth and on volume deviation. No FE or homogenization analysis of any predicted cell is reported, and no failure count (e.g. disconnected or hollow predictions) is given; Figs. 14-16 show selected slices.

P00648 · Brown et al. (2022)

Deep reinforcement learning for engineering design through topology optimization of elementally discretized design domains. Materials & Design 218 (2022) 110672 (Elsevier; issue dated June 2022; gold open access, CC BY 4.0 from 2022-04-16 per Crossref). The pool title carries typographic quotation marks from the publisher metadata. DOI 10.1016/j.matdes.2022.110672. Stratum: none; access: abstract only; stage: adjudicated.

Primary outcome (paraphrase). Not coded (abstract only). The abstract says a deep RL agent that removes elements one at a time produced designs with similar or better compliance than traditional gradient-based TO across several load cases; no numbers and no speed claim. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no marker). no full text

2. **R2: na** (no marker). no full text

3. **R3: na** (no marker). no full text

4. **R4: na** (no marker). no full text

P00663 · Eigel et al. (2022)

Topology Optimisation under Uncertainties with Neural Networks. Algorithms 15(7), 241 (MDPI; published 12 July 2022; open access, CC BY). DOI 10.3390/a15070241. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). Deterministic 2D bridge, nominal load: inserting CNN-predicted gradient steps every 55 iterations cut convergence from 538 to 441 iterations, with lower compliance (1148.66 vs 1173.22) and run time 4 min 05 s vs 4 min 43 s. **Location:** Sec. 4.1.3, Table 1 (p. 22): first quantitative comparison with the classical optimizer (iterations, J, compliance); the matching times are in Table 4 (p. 28). The abstract, conclusion and introduction compare only qualitatively

1. **R1: pass** (R1-matched). Tables 1 and 4: both optimizers solve the same problem with the same volume constraint (m = 0.4) and the same convergence criterion (Eq. A1), and the CNN-assisted design has lower J and compliance at the nominal load, so the time gain (4 min 05 s vs 4 min 43 s) does not come with worse quality. Achieved volumes are not reported. For two of five loads (Table 4) the CNN design is about 0.6% worse in compliance [derived] while faster; a verifier who weighs those cases may choose partial.

2. **R2: pass** (R2-none-established). Sec. 4 and App. B: the baseline is the authors' own phase-field optimizer (refs [1,2]) in the same FEniCS code, and the NN steps are inserted into the same loop, so implementation and hardware are shared [inferred]; hardware and linear solver are not stated. The mesh has up to about 15,000 vertices [$\approx3\times10^4$ DOFs, derived], so (c) does not apply; (b) not observed (no figure in the abstract); (d) not applicable (convergence-based stopping). A per-iteration time is derivable (283 s ÷ 538 ≈ 0.53 s, adaptive mesh [derived]), but the phase-field gradient flow differs from a SIMP strong baseline, so (e) is hard to apply. Unsure, so pass.

3. **R3: fail** (R3-no-cost). Sec. 4.1.1 and 4.2.1: the networks are trained on sample paths of the classical optimizer (S runs with random load rotations, up to 500 iterations each), but S, the data-generation time and the training time are not reported, and no break-even is given. The claim is not limited to per-query use. For C3: the saving is about 38 s per run at the nominal load (Table 4) [derived], so each training path (about one classical run [inferred]) needs roughly 7 uses to pay back [derived].

4. **R4: partial** (R4-no-failure-rate). Tables 1-4: J and compliance come from the FE state solve of the converged designs, which continue with classical steps after each NN step [inferred]. Achieved volume is not reported. Topology collapse for an unsuitable insertion interval c_m or for loads far from the training data is described only qualitatively (Sec. 4.1.4, 4.4), with no counts. The extra loads are a hand-picked selection of extreme cases (three in Sec. 4.1.4 and Table 3; a fourth appears only in Table 4).

P00676 · Suresh (2021)

Length Scale Control in Topology Optimization using Fourier Enhanced Neural Networks. arXiv preprint (Department of Mechanical Engineering, University of Wisconsin-Madison). DOI 10.48550/arxiv.2109.01861. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On four 60x30 benchmarks, Fourier-TOuNN with length-scale bounds (lmin = 6, lmax = 30) reports lower compliance than the 88-line SIMP code in every case (43.39 vs 48.45, 48.81 vs 55.66, 71.2 vs 86.9, 12.23 vs 14.83; Fig. 9). **Location:** Sec. 5.1 (p. 10) and Fig. 9 (p. 11): the first quantitative comparison with a classical method; the abstract, introduction and conclusion make none. The text emphasizes the compliance result (§5.1 rule 4); the FE-iteration counts in Fig. 9 and the run-time remarks of Sec. 5.1 are recorded as alternatives.

1. **R1: partial** (R1-loose). Stratum Q, so R1 applies as written (PROTOCOL §11, 2026-09-24); no time is compared, so no fail marker applies. Sec. 5.1 and Fig. 9 (pp. 10-11): all methods use the same 60x30 mesh, the same target volume fractions and the same CPU. Matching is loose: Fourier-TOuNN's volume is only a growing penalty (Eq. 8, Table 1) and its attained volume is never reported; it runs 151 FE iterations against SIMP's 92-131; and the SIMP penalization for Fig. 9 is not stated, while Fourier-TOuNN's continuation reaches about p = 4.0 at iteration 151 [derived from Table 1], so the printed compliances may not be evaluated at the same penalization. Fourier-TOuNN also carries length-scale bounds that SIMP does not.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no baseline and gives no figures, so (b) is not observed. For the record: the baseline is the published 88-line SIMP code with filter radius 1.4 [50], and all methods ran on the same i7-6700 CPU (Sec. 5, p. 10).

3. **R3: na** (R3-na-per-problem). Stratum Q (PROTOCOL §11): R3 is na. It would be na in any case: a fresh neural network is optimized from scratch for each design case, with no offline training or reusable pretrained model (Algorithm 1, p. 9); this per-problem cost is already inside the reported 4-6 s total time (Section 5.1).

4. **R4: partial** (R4-no-constraint-report). The printed compliance comes from the FE model on the 60x30 mesh (Sec. 3.2, Eq. 7; Fig. 9) [inferred: final-iteration objective], not from an image metric or a surrogate. But the attained volume fraction of the penalty-constrained Fourier-TOuNN designs is not reported for any case, the high-resolution topologies displayed (15x15 samples per element, Sec. 5) are not re-analysed, and disconnected topologies (Sec. 5.5, p. 14) are described without counts: FE-verified, with constraint violations and failures unreported.

P00698 · Kong et al. (2024)

Novel Artificial Neural Network Aided Structural Topology Optimization. Applied Sciences 14(23) (2024) 11416 (MDPI; open access, CC BY 4.0; published 8 Dec 2024). DOI 10.3390/app142311416. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). FE analysis takes over 90% of SIMP run time, but a smaller share in the partitioned-network method: 40% per the abstract, 17-65% across 60×60 to 480×480 meshes in Fig. 14. No absolute run times are reported. **Location:** Abstract (§5.1 rule 1: quantitative comparison with SIMP); data in Sec. 5.4, Figs. 13-14, pp. 18-19

1. **R1: partial** (R1-loose). Sec. 4-5: the cost claim rests on shares of run time (Figs. 13-14) with no absolute times, and the timed runs are not linked to a stated design case [the 3×3-partition L-beams of Table 1 inferred]. Quality is reported separately as unlabelled objective values within about ±9% of SIMP (Tables 1-3 [derived]); volume fractions are not reported, and SIMP stops after a fixed 20 iterations. Matching is therefore loose. A stricter reading (no measured speed at all) gives fail.

2. **R2: pass** (R2-none-established). Sec. 2.1, 4 and 5.4: the SIMP implementation, solver type and hardware are not stated, and no absolute time exists, so (a), (b), (c) and (e) cannot be derived. (d) is not established: SIMP uses 20 iterations while most examples converge within 10 to 20 steps (Sec. 5.1), and both methods share the budget. An FE share of 90-99% is typical of SIMP codes whose cost is dominated by the linear solve, so no inefficiency is evident [inferred]. Unsure, so pass.

3. **R3: partial** (R3-no-amortization). The network is trained offline on random SIMP runs and reused across examples (Sec. 3.2; Conclusion) [reuse inferred]. Training time, dataset size and break-even are not reported; the only cost figure is about 30 min for a process of generating randomly partitioned TO iterations (Sec. 3.3), which may be the data-generation cost [inferred]. The efficiency claim covers the whole design process (abstract), not per-query inference. If the 30 min is not read as a data cost, R3 = fail (R3-no-cost).

4. **R4: partial** (R4-no-failure-rate). Tables 1-6: an objective value is given for every network and SIMP design, presumably from FE analysis of the returned design [inferred; the quantity and its computation are not stated]. Volume fractions and constraint satisfaction of the network designs are not reported, and disconnections or failed runs are not counted; the 25 Appendix A cases show images only.

P00705 · Ullah et al. (2024)

A Machine Learning Approach for Mechanical Component Design Based on Topology Optimization Considering the Restrictions of Additive Manufacturing. Journal of Manufacturing and Materials Processing 8(5), 220 (MDPI; published 1 October 2024; CC BY 4.0). DOI 10.3390/jmmp8050220. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). A Pix2Pix cGAN replaces iterative ToPy SIMP runs (about 90 s each) with millisecond generation for a parameterized 100×100 cantilever; up to 91% of test outputs are artefact-free, judged only by pixel comparison with the TO images. **Location:** Abstract (eliminating the iterative TO steps; 91% reliability), Sec. 1 and Sec. 2.2 p. 6 (90 s per ToPy run vs milliseconds); 91% also in Sec. 3.1 and Sec. 6

1. **R1: fail** (R1-unmatched; R1-no-quality (image-only)). Secs. 1, 2.2, 3.1 and 3.2: millisecond ML generation is set against complete 70-iteration ToPy runs (about 90 s), while the ML outputs are measurably worse: 9% (Models 3-4) to over 50% (Model 6) of test outputs have artefacts or unconnected structures. Quality is compared only by pixel deviation, with no compliance evaluation (Sec. 3.2), and no equal-quality or equal-time matching is attempted.

2. **R2: partial** (R2-inefficient-impl [inferred]; R2(e) re-test candidate). Secs. 2.1-2.2: ToPy (Python SIMP, solver not stated) with a fixed 70 iterations took about 90 s per 100×100 design on an unnamed 12-core CPU at 25% utilization, about 1.29 s per iteration [derived]. Our strong baseline's median per-iteration time on a larger 256×256 cantilever (6.5 times more elements) is 0.336 s with PARDISO (retest/logs/g2_benchmark_2026-09-24.log), already 3.8 times faster [derived]; at 100×100 the gap is likely at least 10 times [inferred], so R2(e) needs our re-test at this size. (b), (c) and (d) are not established: about 2×10^4 DOFs, and 70 fixed iterations with a stated change criterion.

3. **R3: partial** (R3-no-training-cost; R3-no-amortization). Sec. 2.2: data generation is reported (250 h serial, 62.5 h with four concurrent runs, for 10,000 pairs) and called a one-time cost, but training time is not reported and no break-even is given. For C3: the data alone cost the CPU time of 10,000 classical runs, so the model must serve at least 10,000 designs to recoup it, before training cost [derived].

4. **R4: fail** (R4-image-metric). Sec. 3.2: validation is limited to pixel-by-pixel comparison with the ToPy images plus visual inspection, and the authors state that FEM validation was not performed. Artefact and disconnection counts are reported (Sec. 3.1), but no returned design is evaluated for compliance or volume.

P00707 · Xing and Tong (2025)

A stochastic gradient online learning and prediction method for accelerating structural topology optimization using recurrent neural network. Engineering Structures 338 (September 2025) 120507 (Crossref). Hybrid open access: Crossref lists a CC BY 4.0 licence for the version of record; OpenAlex and Semantic Scholar list the ScienceDirect page as the only open copy. DOI 10.1016/j.engstruct.2025.120507. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract claims that SGoLap, a one-hidden-layer recurrent network that learns first- and approximated second-order derivative information online and skips FE and sensitivity steps, saves up to 99.3% (2D) and 90.9% (3D) of total computing time. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P00709 · Gao et al. (2025)

On demand design of two-dimensional phononic crystal bandgap based on machine learning and multi-objective topology optimization. Mechanical Systems and Signal Processing 242 (issue dated January 2026) 113670 (Crossref; published online 2025-11-26 per the pool record). Hybrid open access, CC BY, per OpenAlex and Semantic Scholar; Crossref licence metadata lists only Elsevier text-mining and STM policy URLs. DOI 10.1016/j.ymssp.2025.113670. Stratum: none; access: abstract only; stage: adjudicated.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: VAE, MLP and twin networks predict bandgaps and inverse designs of 2D phononic crystals with five-fold cross-validated accuracy; a twin-network loss balances weight and bandgap targets. No classical comparison. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P00712 · Kallioras and Lagaros (2020)

DL-Scale: Deep Learning for model upgrading in topology optimization. Procedia Manufacturing 44 (2020) 433-440 (conference proceedings; diamond OA, CC BY-NC-ND per OpenAlex). DOI 10.1016/j.promfg.2020.02.273. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract claims that a deep belief network, trained once on a reference problem and applied with SIMP over coarser identical sub-problems, vastly reduces the SIMP iterations needed on fine meshes; no numbers are given. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P00740 · Keshavarzzadeh et al. (2021)

Image-Based Multiresolution Topology Optimization Using Deep Disjunctive Normal Shape Model. Computer-Aided Design 130 (2021) 102947 (received 23 Sep 2019, accepted 20 Sep 2020; online 2020-09-28 per OpenAlex). Crossref lists an Elsevier open-archive licence for the accepted manuscript from 2021-09-29; OpenAlex labels it bronze open access. DOI 10.1016/j.cad.2020.102947. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). 3D cantilever (40x20x4 elements): producing 10,000 high-resolution designs with DDNSM warm starts, counting data generation and training, takes 49.50 h versus 92.00 h for direct SIMP, about half. **Location:** Conclusion (Sec. 5), first quantitative classical comparison there, backed by Sec. 4.3 and Table 13 (p. 23). The abstract compares only qualitatively

1. **R1: pass** (R1-matched [inferred]). Sec. 4.3, Table 13 and Fig. 31 (p. 23): both pipelines end with the same SIMP code and stopping rule, because DDNSM images serve only as initial guesses [inferred: same code, same volume constraint], and the 49.50 h total counts that refinement (166,129.24 s). The refined designs are compared by FE with the direct designs: the Case 6 e_Cfin values on the rendered Fig. 31 lie roughly between 1e-4 and 5e-2 [read from the plot]. The stopping rule is not stated, but both arms share it; the plot suggests a 400-iteration cap on some runs [inferred]. Not the 'loose' case: matching comes from the shared optimizer. The 27.5x no-refinement variant (alternative 2) would fail R1, but it is not the emphasized claim.

2. **R2: pass** (R2-none-established). p. 14 and Sec. 4.3: the baseline is the 88-line code (2D) and the Liu-Tovar 3D code (refs [70], [79]), vectorized educational MATLAB codes [recalled, not verified in-session], run sequentially on an i7-5930K; DDNSM training and evaluation run on K20 GPUs, but the warm-started SIMP also runs sequentially on CPU, and the authors state the two computers are similar in power (p. 14). 12,915 DOF is below 1e5 (p. 20), so (c) does not apply; (b) not observed (no abstract figure); (d) not applicable (convergence-based stopping). Pointer for (e): about 0.218 s per SIMP iteration at 12,915 DOF [derived: 331,213.11 s / 10,000 / 152.08]; 0.066 s at 64x64 and 0.59 s at 200x200 [derived]. Needs a re-test.

3. **R3: pass** (R3-full-accounting). Tables 6, 9, 13 and 15 add dataset generation and network training to the DDNSM totals (pp. 15, 17, 23-24), and break-even counts against direct SIMP are given in FE-cost units (Eqs. 19, 28, 30; Table 5). Caveat: the 3D break-even is miscomputed. Eq. 30, 4.1365e5 + (76.28 x 8)n = (152.08 x 8)n, gives n = 682, not the stated 85 [derived]; the wall-clock break-even from the p. 23 times is about 731 designs [derived].

4. **R4: pass** (R4-FE-verified). Figs. 10-31 and Tables 1-4, 8, 10-12, 14: every test design (400 L-bracket, 200 heat-sink, 200 3D) is used as a SIMP initial guess, and the compliance of the initial guess and of the refined design is computed by FE against the direct design (e_Cini, e_Cfin, p. 10). The share of samples where the warm start did not help is reported (RoS in iterations, e.g. 83.5% success for 3D Case 6, Table 12). The returned designs are SIMP outputs, so the volume constraint is enforced by the optimizer [inferred].

P00755 · Yu et al. (2023)

Artificial neural network supported design of a lattice-based artificial spinal disc for restoring patient-specific anisotropic behaviors. Computers in Biology and Medicine 152 (2023) 106475 (available online 23 Dec 2022; open access, CC BY 4.0). DOI 10.1016/j.compbiomed.2022.106475. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). With the ANN replacing beam FE analysis as the GA fitness evaluator, one lattice-disc topology optimization finishes in about 40 s, versus an estimated 24 h for the same optimization driven by FE analysis. **Location:** Sec. 5.2, p. 10 (first quantitative comparison with a classical method; the abstract, introduction and conclusion claim efficiency only qualitatively)

1. **R1: fail** (R1-no-quality). Sec. 5.2, p. 10: the 40 s ANN-driven optimization is set against an FE-driven optimization that 'will take' about 24 h, an estimate with no reported run, so no classical design or objective exists to compare with. The ANN designs are checked by FE only against the IVD target stiffnesses and against the ANN's own predictions for the same design (Figs. 9, 12, 14), not against an FE-driven GA result at matched quality or matched budget. A speed claim with no quality comparison is a §5.3 R1 fail marker.

2. **R2: pass** (R2-none-established). Sec. 3 and 5.2: both methods would use the same GA settings [inferred] and the same laptop-class CPU (i7-8550U), and no GPU is used, so (c) does not apply; (b) is not observed (no abstract figure). The 24 h estimate has no stated basis: if it assumes all 25,000 evaluations of Table 2 [derived], it implies about 3.5 s per six-load-case beam analysis [inferred], and Fig. 8 suggests that 100-150 of the 500 generations would suffice [read from plot]. But the same generation count applies to the 40 s run, so the ratio does not depend on it, and (d) is not established. The beam-model size is not stated, so efficiency cannot be judged. Unsure, so pass.

3. **R3: partial** (R3-no-data-cost). Training times are reported (about 17 and 23 min per library, Sec. 4.2-4.3), but the cost of the 600,000 FE-analysed training designs per library (Sec. 3.1.4) is not, and no break-even is given; Sec. 5.3 only argues that data generation is a one-time cost. [derived] One library's 600,000 FE analyses equal 24 times the at most 25,000 fitness evaluations of one GA run (population 50 $\times$ 500 generations, Table 2), so break-even needs at least about 24 designs per library, before training time.

4. **R4: pass** (R4-FE-verified). Sec. 3.3 (p. 7) and Figs. 9, 12, 14: every optimized design is re-analysed with the FE beam model ('Optimized true') and plotted next to the ANN prediction; all six runs are shown, including the Library A designs that miss the torsional targets (Fig. 9; Sec. 5.1, p. 8). The problem has no volume constraint, and the cubic unit cells guarantee connectivity between grid-units by construction (Sec. 3.1.2, p. 4); member sizes are checked against the printer resolution (Sec. 5.1). Whether the post-sizing stiffnesses of Fig. 13 come from FE or from the uniform scaling rule is not stated. Unsure, so pass.

P00786 · Luo et al. (2025)

Multi-scale topology optimization of multi-material heterogeneous microstructures based on deep learning. Journal of Computational Design and Engineering 12(8) 127-153 (advance access 16 July 2025; CC BY-NC 4.0). DOI 10.1093/jcde/qwaf064. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The MLP surrogate returns a microstructure's homogenized elasticity tensor and volume fraction about 500× faster than numerical homogenization ($\approx 6\times10^{-4}$ s vs ≈0.32-0.34 s on five example microstructures), with MAE below 0.01. **Location:** Sec. 2.3, p. 134, and Table 2, p. 135 (first quantitative comparison with a classical method, in the main text; the abstract, introduction and conclusions state the cost saving only qualitatively)

1. **R1: pass** (R1-matched-per-evaluation). Sec. 2.3 (p. 134) and Table 2 (p. 135): the primary outcome is per property evaluation, and time and accuracy are reported for the same five microstructures (6)-(10); the surrogate outputs lie within their reported error of the HC result (MAE 0.0011-0.0066, all $<$ 0.01). That is the §5.3 R1 pass condition at the level of the claim, as in P00149. No fail marker holds: a quality comparison exists and no post-processing is omitted from the timed step. Caveats that do not concern matching: no classical multiscale TO is run or timed (claim scope), and the five examples are illustrative picks from Fig. 3B that may lie on the database grid [inferred] (see R4).

2. **R2: pass** (R2-none-established). Sec. 2.3, Table 2: HC implementation, solver and timing hardware are not stated; both methods ran on a CPU workstation [inferred], so (c) does not apply, and the problem has $\approx 2\times10^4$ DOFs [derived], below 10^5. The abstract gives no baseline figure (b); no iteration count applies (d); no per-iteration classical time exists (e). 0.32-0.34 s for one 100×100 homogenization shows no evident inefficiency. Unsure, so pass.

3. **R3: partial** (R3-no-data-cost). Sec. 2.3: training time is reported (131 min, final model; hyperparameter search not timed), but the cost of homogenizing the 156,000 training microstructures is not [inferred ≈14 h at 0.327 s each]. No break-even is given; the introduction argues only qualitatively that the one-off data and training cost is repaid by reuse. C3 [derived]: (51,000 s + 7,860 s) ÷ (0.327 − 0.0006) s $\approx 1.8\times10^5$ property evaluations, i.e. ≈350 iterations at 32×16 cells or ≈8 runs of the 40×4, 135-iteration shape-matching example.

4. **R4: partial** (R4-no-failure-rate). Table 2: surrogate outputs are checked against FE homogenization, but only for five illustrative microstructures; no held-out error is reported beyond the 2.2×10^{-3} loss value (Sec. 2.3). Design objectives (e.g. J = 16.676, Sec. 4.1.2; 19.455, 21.687, 17.580, Sec. 4.1.4) come from the macroscale FE with surrogate-predicted properties; the reconstructed structures are not re-analysed [inferred]. Volume is a soft penalty and final fractions are only plotted (Fig. 12A). One unreasonable design is noted qualitatively (Fig. 14(6)) and isolated single-line boundary microstructures

are described (Sec. 4.1.4); no failure counts. A stricter reading (surrogate-predicted, selected examples) gives fail.

P00808 · Zhang et al. (2026)

A Geometric-Enhanced Neural Network Method for Scalable and High-Resolution Topology Optimization. Symmetry (MDPI), vol. 18 (issue 3 per Crossref), art. 537; published 2026-03-21; open access, CC BY. DOI 10.3390/sym18030537. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On a 2D simply supported beam (160×80 mesh), GeNN reaches a lower compliance than SIMP with a density filter (112,737.99 vs 115,819.95) in 32.55 s instead of 61.36 s, with 236 instead of 400 iterations. **Location:** Section 4.3 'Comparison with the Traditional Method', text and Table 1 (p. 13), the first comparison in the section the article dedicates to the classical baseline; the abstract and conclusion (Sec. 7, point 2) state the same comparison only qualitatively. Chosen under §5.1 rules 2 and 4 over the earlier checkerboard-index numbers of Sec. 4.1 (alternative 2)

1. **R1: partial** (R1-loose). Table 1 and Sec. 4.3 (p. 13): GeNN is both faster (32.55 vs 61.36 s) and lower in compliance than SIMP, so the ML side does not buy speed with worse quality, and no fail marker holds. Matching is loose: SIMP's 400 iterations (500 in Table 2) are reported without any stopping rule, the protocol's own partial example (only GeNN's rule, Eq. 13, is given); the achieved volume fractions are not reported, and GeNN enforces volume only through a quadratic penalty (Eq. 9, p. 8), so equal volume on the physical field is not shown.

2. **R2: pass** (R2-none-established). Sec. 4.3 and Table 1: the 2D baseline is SIMP with a density filter; its solver and the hardware of both methods are not stated (the PARDISO solver is named only for the 3D FEA, p. 14). The abstract gives no baseline number (b). About 2.6×10^4 DOFs [derived] is below 10^5, so (c) cannot apply. The 400 iterations are a round number with no stated stopping rule, but no SIMP convergence criterion or plot exists, so (d) is not established. Per-iteration times [derived]: 0.153 s (SIMP) vs 0.138 s (GeNN) at 160×80; (a)/(e) need the re-test. Unstated hardware is a reporting gap, not an established R2 case; unsure means pass. Pointer: in the 3D alternative (Table 2) SIMP needs 1.456 s per iteration against GeNN's 0.662 s [derived], although GeNN also solves the FE system each iteration.

3. **R3: na** (R3-na-per-problem). Section 3.1 and 3.3: GeNN parameters (ARBF centers/shapes and MLP weights) are randomly initialized and optimized from scratch for each new design problem via Adam, with no labeled dataset and no offline pretraining stage; Section 4.2 shows the only 'reuse' of a trained network is resampling the same already-solved field at higher resolution, not generalizing to a new problem. Learning cost is therefore per-problem and already inside the R1/R2 timing comparison, consistent with the protocol's neural-reparameterization na case.

4. **R4: partial** (R4-no-failure-rate). Compliance is computed via a shared FEA pipeline for both methods (Section 3.2, Eq. 8), i.e., FE-verified rather than an image-similarity proxy. Volume-fraction accuracy is explicitly checked elsewhere in the article (Section 5.2: achieved 0.1998/0.2301/0.1998 against targets 0.20/0.23/0.20), but Tables 1-2 (the primary-outcome comparison) do not restate the achieved volume fraction for either method, and no infeasibility, disconnection or failure-rate check is reported for any of the compared cases.

P00822 · Pereira and Driemeier (2024)

Wasserstein generative adversarial networks for topology optimization. Structures 67 (2024) 106924 (Elsevier; issue dated September 2024; hybrid open access with a CC BY-NC-ND licence per OpenAlex and Semantic Scholar; Crossref lists only text-mining licences). DOI 10.1016/j.istruc.2024.106924. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract says WGANs trained on a TO dataset map physical fields such as von Mises stress to final optimized layouts, removing the iterative algorithm; designs are assessed with metrics, but no classical comparison or number is given. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no marker). no full text

2. **R2: na** (no marker). no full text

3. **R3: na** (no marker). no full text

4. **R4: na** (no marker). no full text

P00827 · Black and Najafi (2024)

Stress-constrained optimization of multiscale structures with parameterized microarchitectures using machine learning. Structural and Multidisciplinary Optimization 67:102 (2024) (Springer; published online 8 June 2024; open access, CC BY 4.0). DOI 10.1007/s00158-024-03821-y. Stratum: speed; access: full text; stage: agreed.

Primary outcome (paraphrase). The two neural-network surrogates evaluate homogenized properties and microstructural stress amplification 1,000-10,000× faster than fullscale FE or successive unit-cell homogenization (about 10^4 on one core), predicting maximum microstructural stress within 10-15% of fullscale FE. **Location:** Sec. 3, p. 11, and the Table 2 note, p. 12 (first quantitative comparison with a classical method, in the main text; the abstract, introduction and conclusion compare only qualitatively; §5.1 rule 2)

1. **R1: partial** (R1-loose). Sec. 3 and Table 2: the speed claim is per evaluation, and its accuracy is reported on four pedagogical cases: the DNN stress error (0.079-0.176) is above the second-order homogenization it replaces (0.046-0.122) in all four and above first-order homogenization in two. No raw times, no optimization run with classical homogenization, and no time- or budget-matched comparison are reported.

2. **R2: pass** (R2-none-established). Table 2 note: the speed-up was measured on a single core; the FE and homogenization implementation and solver are not stated. No GPU is involved [inferred], so (c) does not apply; (b) not observed (no figure in the abstract); (d) and (e) are not derivable (no iteration count or per-iteration time). Unsure, so pass.

3. **R3: partial** (R3-no-training-cost). p. 11: data generation is reported (about 1 h on four cores), but training time is not, and no break-even or per-query limitation is given. For C3: the data cost is about 4 core-hours [derived: 1 h × 4 cores]; without a training time or any per-evaluation time, a break-even cannot be computed from the article.

4. **R4: pass** (R4-FE-verified). Sec. 5 and Table 4: every optimized design is resolved on the fullscale mesh and its von Mises stress recomputed; net volume and the fraction of yielded microstructures are reported for all designs, and the non-converged HS-truss case is disclosed (Sec. 5.1). Surrogate accuracy is checked against fullscale FE (Table 2) and on validation sets (Figs. 10-11).

P00842 · Doosti et al. (2021)

Topology Optimization via Frequency Tuning of Neural Design Representations. Symposium on Computational Fabrication (SCF '21), October 28-29, 2021, Virtual Event, USA (ACM). DOI 10.1145/3485114.3485124. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On three 2D benchmarks (MBB 300×100, roof 180×180, bridge 250×125), ENDR designs have lower compliance than the voxel-based density method (285.7 vs 316.7; 2.731 vs 2.905; 9.732 vs 9.785), using 5000 iterations against about 300. **Location:** Sec. 5.4 (p. 6) and Fig. 6 (p. 7). The abstract (p. 1) and conclusion (Sec. 7, p. 7) compare with standard methods only qualitatively ('state-of-the-art performance'; 'outperform or be on par with standard FEM-based solutions in terms of stiffness'); chosen under §5.1 rules 2 and 4 as the comparison the article emphasizes, over the iteration counts of the same paragraph, which the article presents as a cost limitation (alternative 1)

1. **R1: partial** (R1-loose). Sec. 5.4 (p. 6) and Fig. 6 (p. 7): R1 applies as written in stratum Q (PROTOCOL §11). Both methods solve the same three problems at the same volume budgets, and compliances are printed; no fail marker applies, because no time is compared. Matching is loose: ENDR gets 5000 iterations against the voxel method's ~300; ENDR's volume is a soft penalty 'generally' within 1e-4 (Sec. 4.3, p. 4), with no per-case volume reported; and the voxel method's stopping rule for Fig. 6 is not stated (Eq. 5 covers only the upsampling runs of Sec. 5.3). This is the §5.3 R1 partial case.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) names no baseline and gives no numbers ('state-of-the-art performance'), so it cannot report a weaker baseline than Sec. 5.4 and Fig. 6; (b) does not apply. For reference: the

voxel-based baseline follows Andreassen et al. 2011 with filter radius 1 and converges in about 300 iterations (Sec. 5.4); the authors' own solves use sparse Cholesky (2D) and multigrid-preconditioned CG (3D) (Sec. 5.6).

3. **R3: na** (R3-na-quality-only). Stratum Q (PROTOCOL §11): R3 concerns speed claims and is na. It would be na in any case: ENDR is a per-instance, self-supervised optimization with no dataset and no offline training (Sec. 4.2), the §5.3 per-problem case.

4. **R4: partial** (R4-no-failure-rate). The compliance of each returned ENDR design is computed with the same FEM model used during optimization (Sec. 3.1, 4.2, pp. 2-3; J values printed in Fig. 6, p. 7), not with an image metric or a surrogate. Volume satisfaction is stated only as a general bound (violation 'generally' below 1e-4, Sec. 4.3, p. 4), not per case, and no failure, disconnection or infeasibility count is reported for the comparison or upsampling results.

P00860 · Garrelts et al. (2021)

AI-Based Topology Optimization of Freehand Sketches. Procedia CIRP 104 (2021) 1316-1321 (Crossref; conference proceedings, CC BY 4.0 per Crossref and OpenAlex). DOI 10.1016/j.procir.2021.11.221. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: a neural network reads photographed design sketches and generates lightweight structures matching the sketched load cases; the abstract states no comparison with a classical method. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P00875 · Ma et al. (2023)

Learning topology optimization process via convolutional long-short-term memory autoencoder-decoder. International Journal for Numerical Methods in Engineering 124(11) (2023) 2571-2588 (Crossref; hybrid open access, CC BY 4.0 per Crossref and OpenAlex). DOI 10.1002/nme.7221. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: a ConvLSTM encoder-decoder, fed the first SIMP iterations, predicts the rest of the optimization path, claiming a large computation-cost reduction for a small loss in design performance. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P00890 · Bujny et al. (2023)

Learning hyperparameter predictors for similarity-based multidisciplinary topology optimization. Scientific Reports 13 (2023) 14856 (Nature Portfolio; published 8 September 2023; open access, CC BY 4.0). DOI 10.1038/s41598-023-42009-0. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). A degree-5 polynomial regression predicts the whole dissimilarity-versus-scaling-factor curve from one standard TO run, which the authors count as 12 times less computation than sampling the curve with 12 similarity-based TO runs. **Location:** Section 'Regression model', p. 7. Chosen under §5.1 rule 4: the abstract, introduction and conclusion state the cost saving only qualitatively, and in the main text the cost statement follows the test-set accuracy ($R^2 = 0.96$) in the same paragraph; the article emphasizes the cost saving

1. **R1: partial** (R1-loose). p. 7 and Fig. 2: the 12x is a run count (1 standard TO run vs 12 sampled ESM runs), not a measured time, and quality is reported (test-set R^2 0.96, RMSE 0.027; Figs. 16 and 19). Matching is loose: the predicted curve is an approximation of the sampled one, the stopping rules of the TO runs are not stated, and the extra ESM run needed to produce the design at the chosen p is not counted (design-level ratio 6x [derived]). A verifier who reads that uncounted run as post-processing outside the timing may choose fail.

2. **R2: pass** (R2-none-established). pp. 5-7, Tables 2-4 and the Data availability statement (p. 16): both arms run the same TO code per run (ESM in SIMP for the 2D data, HCA for crash and hood), and the claim is a run-count ratio, so the solver and hardware of each run cancel; no run time, solver or hardware is stated for any run, and the ML step is a polynomial evaluation. None of (a)-(e) is established: the abstract gives no number (b), no GPU asymmetry is described (c), and no iteration count or per-iteration time exists (d, e). Unsure, so pass. Baseline-strength note, not an R2 marker: the comparator samples p on a uniform 12-point grid; because s should fall as p rises (p. 10), a bracketing search could hit one target similarity in about 4 runs at the same resolution [derived: ceil(log2 12)], although the full s(p) curve that the article emphasizes (p. 7) needs all 12 samples.

3. **R3: partial** (R3-no-amortization). p. 5 and 7: the data cost is given as a run count (7800 TO runs of the 100×100 2D cantilever); run times and training time are not reported, and no break-even is given; the claim is not limited to per-query use. For C3: counting runs, the saving is 10 TO runs per use (12 vs 2 [derived]), so the 7800 training runs break even after about 780 uses [derived]; in time the break-even is likely far lower, because the training runs are small 2D SIMP runs while the savings accrue on crash or 3D industrial runs [inferred].

4. **R4: partial** (R4-no-failure-rate). p. 7, Figs. 8, 16, 19: predictions are checked against actual similarity-based TO runs (FE-based): the test-set metrics cover 60 held-out pairs, and the Fig. 8 curves are randomly selected (stated rule). The designs themselves come from full TO runs, so volume is enforced by the optimizer [inferred]. But the crash and hood checks are graphical only, and cases where the predicted p misses the target similarity, or where the degree-5 model is non-monotonic near the feature bounds (p. 10), are not counted.

P00918 · Martínez-Martínez et al. (2025)

Accelerating cell topology optimisation by leveraging similarity in the parametric input space. Computer Methods in Applied Mechanics and Engineering, vol. 442, art. 118044 (published online 9 May 2025; open access, CC BY 4.0). DOI 10.1016/j.cma.2025.118044. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Warm-starting cell-level SIMP with the nearest archived optimised cell (energy-based similarity, augmented 15,744-sample dataset) instead of a uniform density cuts the mean SIMP iteration count on the held-out 2D Hook problem by a factor of 2.67 (Table 5). **Location:** Section 4 (Results), Table 5 (p. 17), with the per-cell results of Figs. 16-17, 20-21, 24, 27-28; the abstract and conclusions (Sec. 6, pp. 21-23) state the acceleration only qualitatively. Chosen under §5.1 rules 2 and 4: Table 5 is the article's summary of the acceleration, and the energy-based metric with Dataset Index 3 and augmentation is the configuration the conclusions recommend

1. **R1: partial** (R1-loose). Sec. 3.3-5.3 and Table 5: both arms run the same SIMP code to the same convergence check, so the iteration ratio is taken at convergence for both, and no fail marker holds. Matching is loose: the convergence criterion is never stated numerically (Sec. 2.2, p. 4), the warm start leads to different local optima (Sec. 5.2, p. 20), and the final-compliance match is shown only for two example cells (Figs. 29-30) and qualitatively for the assembled structure (Fig. 31), with no per-cell compliance for the 486 test cells. This is the §5.3 R1 partial case.

2. **R2: pass** (R2-pass-shared-solver). The classical ('standard') and proposed arms use the identical FE/SIMP implementation and (implicitly) identical hardware; only the cell's initial density field differs (Sec. 3.3, Fig. 10). No re-test data, no abstract-vs-results baseline mismatch, no GPU-vs-CPU asymmetry, and no fixed-iteration-cap vs. convergence-based mismatch is present (both use the same convergence-based stopping), so none of R2(a)-(e) is evidenced; per the template convention, unsure/no adverse evidence defaults to pass.

3. **R3: partial** (R3-no-data-cost). The instance-based model has no training phase (Sec. 3.3, p. 10), so there is no training cost. The data cost is reported only as sample counts: 984 cells from macro problems A-C plus an equal number of context-based samples, each needing its own cell TO solve

(Tables 1-2, pp. 7-9; Sec. 3.2.1; p. 22), i.e. 984 dedicated extra cell optimisations at Index 3 [derived]. No time or iteration count is given for building the archive, and no amortization or break-even against the per-cell saving is computed. Costs reported without amortization are the §5.3 R3 partial case; the article does address the data cost, so the fail marker (no training or data cost at all) does not hold.

4. **R4: partial** (R4-no-failure-rate). Every returned cell design in both arms comes from the real FE/SIMP solve (not a surrogate prediction), and the volume fraction is a constraint of that solve [inferred: OC enforces it]. But compliance is reported only for two example cells (Figs. 29-30) and qualitatively for the assembled structure (Fig. 31), no infeasibility or failure count is given, and inter-cell continuity is argued qualitatively (Sec. 5.3, p. 20). The decelerated cells are visible only in the per-cell plots (Figs. 17, 21, 28).

P00983 · Xing et al. (2025)

A function online learning and prediction method for accelerating structural topology optimization and its application to pneumatic soft actuator. Thin-Walled Structures (2025), article 112942; hybrid open access, CC BY per OpenAlex. DOI 10.1016/j.tws.2025.112942. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract claims that FoLap, a one-hidden-layer network that learns and predicts design-function updates online, saves up to 65.1% of total computing time for a soft-actuator cell design versus level-set and moving-isosurface-threshold TO. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P01016 · Ghansiyal et al. (2024)

Designing Porous Structure with Optimized Topology using Machine Learning. Procedia CIRP 125 (2024) 190-195, conference proceedings (Crossref); open access, CC BY-NC-ND 4.0 (Crossref licence metadata; OpenAlex: diamond). DOI 10.1016/j.procir.2024.08.033. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract says machine-learning models performing topology optimization for porous biomedical structures give a large cut in computation time, with performance comparable to conventional methods; no numbers are given. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P01105 · Zhu et al. (2021)

Optimisation of three-dimensional hierarchical structures with tailored lattice metamaterial anisotropy. Materials & Design 210 (2021) 110083 (Elsevier; available online 2 September 2021; open access, CC BY-NC-ND 4.0 per the article's first page). DOI 10.1016/j.matdes.2021.110083. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). MBB beam at volume fraction 0.3: the ANN-surrogate lattice optimization with orthotropic PC cells (PC-A) reaches 30.6% lower compliance than solid-void SIMP (p = 3) and 13.0% lower than quasi-isotropic PC lattices, on the same Nastran platform. **Location:** Sec. 6 Conclusions, p. 17 (first quantitative comparison with a classical method; the abstract's comparison is qualitative, §5.1 rule 2); the same values are plotted in Fig. 16, p. 13

1. **R1: partial** (R1-loose). Sec. 5 and Figs. 12, 16, 19 (pp. 12-15): R1 applies as written in stratum Q (PROTOCOL §11). All designs share the compliance problem, volume-fraction bound, mesh, filter, MMFD optimizer and convergence criteria, and no time is compared, so no fail marker holds. Matching is loose: the convergence tolerances are not stated (Sec. 2.1, p. 4); achieved volume fractions are not reported, and for the lattice cases the volume comes from the ANN relative-density surrogate rather than the geometry (Eqs. 3.3 and 19; Table B2 RMSE 0.0081-0.0126 [read from table]); the SIMP p = 3 designs keep filter-induced grey regions without post-processing (p. 11). This is the §5.3 R1 partial case.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) claims superiority over 'classical topology optimisation'; the results define it as solid-void SIMP p = 3 (Iso-P3, Sec. 5, p. 9; Sec. 6, p. 17). The results also run SIMP p = 1 and p = 2, which match (p = 2) or beat (p = 1) PC-A (Figs. 12, 16, 19), but the article states that their intermediate densities lack physical meaning and uses them only as indications (p. 10), so they are not a stronger valid baseline hidden by the abstract; (b) is not established. No run time is reported for any method.

3. **R3: na** (R3-na-quality-only). Stratum Q (PROTOCOL §11): R3 concerns speed claims and is na. The efficiency gain of the ANN over direct homogenization is asserted (abstract; Sec. 2.1, 3.3) but never measured. For C3: the homogenization of 165, 330 and 1,327 samples (Sec. 3.1) and the ANN training are not costed; no break-even is given.

4. **R4: partial** (R4-no-failure-rate). Figs. 12, 16 and 19: compliance comes from a Nastran FE analysis of the optimised designs, but of the homogenised model with ANN-predicted properties. The returned lattice CAD model, with radii averaged at strut junctions (p. 14), is not analysed in detail. The homogenization check covers uniform lattices only (Sec. 5.1, Table C1: strain-energy difference below 5% at 8 cells per dimension). Achieved volume fractions and failed runs are not reported. A stricter reading (lattice performance partly surrogate-predicted) gives fail.

P01151 · Honarvar et al. (2026)

Two-scale topology optimization of multiscale structures and materials via deep neural networks for local and global buckling. Computer Methods in Applied Mechanics and Engineering 458 (August 2026) 119038 (Crossref); published online 2026-05-09 per the pool record; hybrid open access, CC BY-NC 4.0 (Crossref licence metadata; Elsevier API core metadata: sponsor Drexel University). DOI 10.1016/j.cma.2026.119038. Stratum: none; access: no access; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text and no abstract available). Only the title is known: deep neural networks are used in two-scale topology optimization of structures and their materials under local and global buckling. **Location:** title only

1. **R1: na** (no-full-text). no full text

2. **R2: na** (no-full-text). no full text

3. **R3: na** (no-full-text). no full text

4. **R4: na** (no-full-text). no full text

P01217 · Parida and Dash (2023)

Enhancing Structural Topology Optimization with Deep Learning: A Stress Field-Based Approach. Advancement in Mechanical Engineering and Technology 6(2) (2023) 24-37 (HBRP Publication), per the PDF header and the Zenodo record (resource type 'Journal article'). The Zenodo record lists CC BY 4.0; the PDF footer reads 'All Rights Reserved'. DOI 10.5281/zenodo.8003313. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On five 32×96 cantilever cases, the Attention Res-U-Net is claimed to match SIMP and Abaqus designs within 5% volume-fraction error in one tenth of their time (per case 3 s + 0.004 s vs 10 s and 20 s). **Location:** Abstract (quantitative comparison with classical methods, §5.1 rule 1); supporting numbers in Table 3 (p. 34), Table 4 and Figs. 9-10 (p. 35); repeated in the Conclusion (p. 36)

1. **R1: fail** (R1-unmatched). Table 4, Table 3 and Figs. 9-10: DL inference (3 s + 0.004 s) is set against complete SIMP (10 s) and Abaqus (20 s) runs, while the DL designs are measurably worse on both counts: they use 0.53-0.54 volume against a 0.5 target (SIMP 0.49-0.51), and their strain energy is higher than SIMP's in all five cases despite the extra material. No matching is attempted, and the SIMP stopping rule is not stated.

2. **R2: partial** (R2-inefficient-impl). p. 34: the SIMP baseline is the original 99-line MATLAB educational code, and the other baseline is Abaqus CAE's general-purpose optimization module; the hardware of the timed runs is not stated for either method. The mesh has about 6.4×10^3 DOFs [derived], so (c) does not apply; (b) not observed (the abstract gives no baseline time); no iteration count is given, so (d) and (e) cannot be derived.

3. **R3: partial** (R3-no-data-cost). p. 32: training time is reported (about 0.9 h), but the cost of the 2,000 Abaqus optimizations is not, and no break-even or per-query limitation is given. For C3 [derived]: at the Table 4 times, data cost about 2,000 × 20 s = 40,000 s plus 3,240 s of training; per-use saving 16.996 s vs Abaqus and 6.996 s vs SIMP, so break-even after about 2,540 and 6,180 uses.

4. **R4: partial** (R4-no-failure-rate). pp. 34-35: the five DL designs of the primary comparison are scored by their achieved VF (Table 3) and by strain energy against SIMP and Abaqus (Fig. 10), which needs an analysis of each returned design [FE inferred; the method is not stated]. Failed or disconnected predictions are neither defined nor counted (the L4 design in Fig. 9 shows a thin member with intermediate densities), and the VF excess over the 0.5 target is presented as a small error. The 600-sample test set is scored only by image metrics (Table 2, Fig. 7) and a VF correlation (Fig. 8); the five load cases are the designated comparison set, 'adopted' without a stated rule.

P01219 · Cheng et al. (2024)

Real-time multi-objective topology optimization method via deep learning. In: Bridge Maintenance, Safety, Management, Digitalization and Sustainability (Jensen, Frangopol & Schmidt, eds.), 2024, ISBN 978-1-032-77040-6, pp. 2090-2097 (chapter header and Crossref). Open access via taylorfrancis.com, CC BY-NC-ND 4.0. DOI 10.1201/9781003483755-246. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On the 40x80 cantilever test set, nearly half (49.75%) of the network's predicted designs beat the MMA reference designs, meaning compliance and p-norm stress over 1% lower with volume within 1%. **Location:** Conclusion (Sec. 5, p. 2097), the first quantitative comparison with a classical method there; the share is Table 1 'OR' (p. 2096), defined in Sec. 4 (p. 2095). The abstract compares with the conventional method only qualitatively

1. **R1: partial** (R1-loose). Stratum Q; R1 applies as written (PROTOCOL §11, 2026-09-24). Sec. 4 (p. 2095), Table 1 (p. 2096): predictions and MMA designs are compared on the same test inputs through compliance, p-norm stress and volume errors. Matching is loose: 'outperform' admits a volume difference of up to 1% (mean volume error 0.84%), the MMA stopping rule and iteration count are not stated (only 'hundreds of iterations'), and the MMA references are beaten on average (Ec -7.50%, p-norm stress -5.08%), which suggests under-converged or locally optimal references [inferred]. 5.9% of predictions have errors above 20%.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 2090) gives no baseline figure and names no weaker baseline; the results compare with the same MMA method (Sec. 3.1, p. 2092; Sec. 4, p. 2095), so (b) is not observed. For the alternative speed outcome: MMA implementation, solver and hardware are not stated; about 6,640 DOFs [derived], so (c) does not apply; (d) and (e) are not established.

3. **R3: na** (R3-na-quality-stratum). The primary outcome is quality only. For C3 and the alternative speed outcome: Sec. 4 states that dataset generation (20,000 MMA runs, Sec. 3.1) and training (136 epochs, Sec. 3.3) are excluded from the reported time, and neither cost is given; no break-even. [derived] At 15.76 s per MMA run the data alone cost about 87.6 h and are recovered after about 20,000 uses. If the speed outcome is taken as primary, R3 = fail (R3-no-cost).

4. **R4: pass** (R4-verified [inferred]). Sec. 4, Table 1, Fig. 4: compliance and p-norm stress of the returned predictions are reported against MMA, which implies FE analysis of the predicted designs [inferred; the procedure is not described]. Volume error is reported (mean 0.84%) and large-error samples are counted (5.9%), with disconnected designs named as one cause. The Fig. 4 examples have no stated selection rule but supplement aggregate metrics.

P01232 · Zhang et al. (2024)

Research on topology optimization of carbon fiber reinforced composite electric vehicle helmet based on deep learning. Journal of Physics: Conference Series 2791 (2024) 012064, CMSDA-2023 proceedings (IOP Publishing; open access, CC BY 4.0 per the article's first page). DOI 10.1088/1742-6596/2791/1/012064. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). A modified U-Net trained on SIMP designs of the flattened helmet grids (30,000 samples in all, split 8:1:1) reproduces the SIMP element densities with a mean-squared-error loss of about 0.005 on the training set and 0.01 on the held-out verification set after 100 epochs. **Location:** Sec. 3.3, p. 7, Eq. 13 and Fig. 10 (first quantitative comparison with classical results, in the main text; §5.1 rule 2). The abstract, introduction and conclusion compare the optimized helmet only with an unoptimized ABS helmet, and the conclusion's speed statements are qualitative

1. **R1: partial** (R1-loose). Sec. 3.1 and 3.3, Fig. 10: predictions and SIMP labels come from the same problem generator (held-out verification split), so the problem settings match, but agreement is measured only as an MSE loss. The compliance and volume fraction of predicted designs are not reported, the SIMP stopping rule is not stated, and no time or FE budget is matched.
2. **R2: na** (R2-na-quality-only). No speed or cost comparison is quantified: no classical time, iteration count or hardware is reported (Sec. 3.1, 3.3), so criteria (a)-(e) cannot apply. The SIMP labels come from a MATLAB code built on the cited 99-line and 88-line educational codes [inferred from p. 2, p. 5 and refs 9-10]. Stratum Q (PROTOCOL §11, 2026-09-24): the abstract (p. 1) names SIMP only as the data generator and gives no baseline figures, so criterion (b) is not observed.
3. **R3: na** (R3-na-no-speed-claim). The only speed statements are qualitative (Sec. 6, p. 12). For C3: neither the cost of the 30,000 SIMP samples (p. 5) nor the training time (100 epochs on three RTX 3090 GPUs, p. 7) is reported; no break-even is given.
4. **R4: fail** (R4-image-metric). Sec. 3.3 and Fig. 10: agreement with SIMP rests only on the MSE loss of element densities. No FE analysis of a predicted 2D design, no volume check and no failure count are reported. The 3D helmet is FE-analysed in ABAQUS (Sec. 5.2) only after the stiffeners were modified by hand to remove checkerboard regions (p. 9, Figs. 15-16), and only against an ordinary ABS helmet.

P01268 · Aroyan et al. (2026)

Application of a Machine Learning-Based Topology Optimization Method on an Automotive B-Pillar for Crashworthy Lightweight Design. MATEC Web of Conferences 422, 03010 (2026); conference paper, CC BY 4.0 per Crossref licence metadata (licence start 2026-09-09). The DOAJ PDF file name refers to EngOpt 2026 [inferred from the file name]. DOI 10.1051/matecconf/202642203010. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract reports voidML cut the B-pillar patch mass from 2.413 to 2.141 kg (11.3%) in 48 iterations and 191 crash analyses, with force-intrusion curves close to the reference; no classical comparator is named. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P01289 · Vulimiri et al. (2026)

Process-aware topology optimization leveraging deep learning surrogate model-based functions for laser powder bed fusion. Additive Manufacturing 118 (issue dated February 2026) 105094 (Crossref; published 2026-01-23 per the pool record). Hybrid open access, CC BY-NC, per OpenAlex and Semantic Scholar; Crossref licence metadata includes CC BY-NC 4.0. DOI 10.1016/j.addma.2026.105094. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: a surrogate trained on elastoplastic layerwise inherent-strain FEA drives distortion-aware LPBF topology optimization; the largest

case, over 400 iterations, finished in 2.5 days versus 4 days for one verification FEA process simulation. **Location:** Abstract and highlights (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P01320 · Ouchaoui et al. (2024)

Topology optimization using artificial intelligence. Incertitudes et fiabilité des systèmes multiphysiques (ISTE OpenScience), vol. 8, no. 2, pp. 38-42 (per the PDF file name and page footers; diamond open access per OpenAlex). DOI 10.21494/iste.op.2024.1231. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Cantilever (120x60 elements, volume fraction 0.4): against the SIMP design, the TopologyGAN design has 2.72x the peak von Mises stress, 5.62x the peak displacement and 8.08x the deformation energy. **Location:** Sec. 3, Table 1 and the paragraph after Fig. 3 (p. 41): the first quantitative comparison with a classical method. The abstract and conclusion compare only qualitatively (the key phrase is from the abstract)

1. **R1: partial** (R1-loose). Sec. 3, Table 1: one TopologyGAN design is compared with one SIMP design, but the volumes differ (weight reduction 46% vs 40%, i.e. 54% vs 60% of the solid weight [derived]), although both nominally target volume fraction 0.4. How the TopologyGAN design was produced (pretrained or retrained model, input resolution, post-processing) is not stated, and the SIMP stopping rule is not given. Quality is reported, but matching is loose.
2. **R2: na** (R2-na-quality-only). Stratum Q: only the SIMP time (54 s) is reported; with no TopologyGAN time there is no speed comparison, so criteria (a)-(e) do not apply. Under the §11 stratum-Q rule (2026-09-24) only criterion (b) could apply, and the abstract (p. 38) gives no baseline figures, so (b) is not observed.
3. **R3: na** (R3-na-no-speed-claim). No quantified speed claim. The abstract and conclusion assert better computing performance without any time; for C3, no training or data-generation cost is mentioned.
4. **R4: partial** (R4-no-constraint-report). Table 1 (p. 41): both returned designs are scored by the same static analysis (peak stress, displacement, deformation energy) [FE re-analysis inferred; the tool is not named], on the smoothed geometries of Fig. 3, whose post-processing is not described. The reported weight reductions (40% SIMP, 46% TopologyGAN, p. 41) leave 60% and 54% of the solid plate of Fig. 2 (p. 40) [derived], against vf = 0.4 (p. 40); this violation is not acknowledged, and failed designs are neither defined nor counted.

P01390 · Venugopal and Anand (2023)

Structural and thermal generative design using reinforcement learning-based search strategy for additive manufacturing. Manufacturing Letters 35 (2023) 564-575 (Elsevier; Crossref type journal-article, issue dated August 2023; version-of-record licence CC BY-NC-ND 4.0 from 2024-02-13 per Crossref). DOI 10.1016/j.mfglet.2023.08.030. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract says an Upper Confidence Bound (UCB) search perturbs SIMP to give diverse compliance and heat-conduction designs with AM filters, and calls the method computationally efficient and scalable; no classical comparison or number is given. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P01517 · Li and Ye (2025)

A Multi-Grid, Single-Mesh Online Learning Framework for Stress-Constrained Topology Optimization Based on Isogeometric Formulation. Computer Modeling in Engineering & Sciences 145(2) (2025) 1665-1688 (Tech Science Press; diamond open access, CC BY 4.0; received 27 Aug 2025, issue published 26 Nov 2025). DOI 10.32604/cmes.2025.072447. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On two 2D stress-constrained L-bracket problems (103×52 control points), FNO online learning cut IGA evaluations from 210 to 150 and total time by about 20% (2540.44 to 2062.78 s; 2429.20 to 2058.63 s) at matching volume and maximum stress. **Location:** Sec. 5.1 and Table 1 (first quantitative comparison with the classical TO-IGA run; the abstract and conclusion compare only qualitatively, and the introduction cites other studies' results only). Sec. 5.2.4 states the Table 1 speed-up as 1.23

1. **R1: pass** (R1-matched). Sec. 5.1, Table 1 and Fig. 10: both methods run the same code, formulation and stopping rule (210 iterations or mean design change below 1e-3) on one machine, and the online-learning designs match the baseline (volume 0.1489 vs 0.1500 and 0.1464 vs 0.1465; max stress 24.9176 vs 25.2465 and 24.9528 vs 24.9757 against a 25 Pa limit). The time comparison is therefore made at matched design quality. No post-processing is applied to either design.

2. **R2: partial** (R2-inefficient-impl). Sec. 5, Table 1: the baseline is the authors' in-house MATLAB isogeometric TO code, the same code the online framework calls. It takes about 12.1 s per iteration at 10,712 DOFs [derived: 2540.44 s ÷ 210], and Sec. 4 names the stress-sensitivity step as the most time-consuming step. This looks slow for the size [inferred]; the article itself cites efficiency-oriented MATLAB IGA-TO implementations ([18], [19]), but no direct comparison is possible. A faster baseline would shrink the saving, because the FNO overhead stays fixed [inferred]. (c) does not apply (at most 4.2×10^4 DOFs [derived]); (b) and (d) are not observed (no abstract figure; both methods share the 210-iteration cap, and Fig. 10 shows the volume still falling until about iteration 170 [read from plot]). Unsure alternative: pass. For scale only: the project's compliance-SIMP strong baseline needs 0.0045 s per iteration at n = 64 with PARDISO (retest/logs/g2_benchmark_2026-09-24.log), a different problem class, so R2(e) cannot be applied.

3. **R3: na** (R3-na-per-problem-learning). Sec. 4 and 5.1: the FNO is trained online on data from the same optimization run (warm-up n0 = 20 IGA iterations, then n1 = 10 IGA and n2 = 5 FNO iterations per block, 150 epochs per training), with no offline dataset or pretraining. Its cost sits inside the total times compared under R1 [inferred: Sec. 5.1 reports the whole run time].

4. **R4: pass** (R4-FE-verified). Tables 1, 2, 4 and 5: volume fraction and maximum von Mises stress are reported for every returned design, with stress plots at the Gauss points; they come from IGA analysis of the final design [inferred from Sec. 4, where the design at the last iteration is evaluated]. Constraint satisfaction is visible for each case, including small overshoots of the stress limit by both methods (e.g. online learning 40.4327 vs 40 Pa, Table 2). All runs are shown; failed runs are neither reported nor discussed, and each case is a single run. Unsure, so pass.

P01813 · Fernando et al. (2026)

Diffusion-based cGAN framework for automated optimization and code-compliant generative design of thin-walled structures. Thin-Walled Structures 231 (2026) 115420 (Elsevier; special issue 'AI-Assisted Thin-Walled Structure Design'; received 15 May 2026, available online 24 July 2026; open access under CC BY 4.0 per the article and Crossref). DOI 10.1016/j.tws.2026.115420. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Generated 512×512 frame layouts reproduce the reference layouts of the conventional TopOpt-plus-sizing pipeline with SSIM up to 0.956 and Dice up to 0.985 (simply supported: SSIM 0.917-0.956; cantilevered: 0.841-0.93), shown for representative test examples. **Location:** Abstract (the only quantitative comparison with the classical pipeline's output; its speed statement is qualitative); numbers in Sec. 4.3, Table 1 and Fig. 13; repeated in Sec. 5 (Conclusions)

1. **R1: partial** (R1-loose). Sec. 4.3, Table 1, Fig. 13: generated and reference layouts share the input parameters (F, S, H), so the comparison is matched by construction, but quality is only image similarity on representative examples. No structural performance (compliance, mass, utilization) of the generated designs is compared with that of the reference designs; the two designs re-analysed in Sec. 4.6 are checked against code limits only.

2. **R2: na** (R2-na-quality-only). The primary outcome is image similarity to the reference layouts, so the efficiency criteria (a)-(e) do not bear on it. For the Table 2 speed alternative, R2 would be partial (the hardware and implementation of the conventional pipeline are not stated); see notes. Under the §11 stratum-Q rule (2026-09-24) only criterion (b) could apply: the abstract (p. 1) gives no baseline figures and names the same conventional iterative pipeline that Table 2 (p. 14) times, so (b) is not observed.
3. **R3: na** (R3-na-quality-stratum). Quality stratum. For C3: the authors state that the Table 2 times exclude dataset generation, hyperparameter tuning and training (Sec. 4.4) and limit the speed claim to online inference; training is reported only in epochs; the 3513 layouts (split into training, validation and test sets) come from the conventional pipeline (Sec. 3.4) [inferred ≈429 h at the Table 2 mean times].
4. **R4: fail** (R4-image-metric). Sec. 4.3, Table 1, Fig. 13: the quality result rests only on image-similarity metrics (SSIM, PSNR, IoU, Dice) for representative examples; no selection rule and no test-set average are given. The frame re-analysis in Sec. 4.6 (Tables 6-8) covers two representative generated designs, checks code limits rather than the reference designs, and reports no failure rate. Table 3 gives pixel metrics against ML baselines only.

P01827 · Carson and Hirschfeld (2026)

TPMS-Optimized Cement Geometries for Carbon-Efficient Structural Performance. Journal of High School Research 3(1) (2026), article 72 on jhsr.net (published 31 Aug 2026; CC BY-NC-ND 4.0 per the article page). DOI 10.70671/dq9am588. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract only: ML surrogates predict the performance of swept TPMS cement geometries (R^2 0.987-0.999), and generated hybrid topologies reach 104% of the best classical TPMS compressive performance. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P01870 · Chu et al. (2024)

Exploring VAE-driven implicit parametric unit cells for multiscale topology optimization. Materials & Design 244 (August 2024) 113087 (Crossref); gold open access, CC BY 4.0 (Crossref licence metadata; Elsevier API core metadata: DEAL agreement). DOI 10.1016/j.matdes.2024.113087. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract describes a VAE with a property regressor, trained on a unit-cell database built by an AAPO topology-optimization method plus a distortion filter, for multiscale thermo-mechanical metamaterial design. Efficiency is asserted, with no classical comparison named. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P01907 · Ignuta-Ciuncanu et al. (2025)

Evolutionary design of conductive pathways using a generative autoencoder. International Communications in Heat and Mass Transfer 166 (2025) 109098 (online 2025-06-02; hybrid OA, CC BY per OpenAlex). DOI 10.1016/j.icheatmasstransfer.2025.109098. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract claims that a VAE coupled with evolutionary search lowers average device temperature by 10% and peak hotspot temperature by up

to 1%, and raises Pareto hypervolume by 13%, compared with unnamed standard methods. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P01960 · N et al. (2025)

Reinforcement learning-based topology optimization for generative designed lightweight structures. MethodsX 15 (Dec 2025) 103539, Elsevier, open access under CC BY 4.0 per the licence statement of the PMC/Europe PMC XML (the first coder had recorded CC BY-NC-ND) (received 24 May 2025, accepted 28 Jul 2025, per PMC). DOI 10.1016/j.mex.2025.103539. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The abstract says the PPO-based design method beats conventional SIMP and level-set approaches on the ToD benchmark, with up to 40% lower weight at maintained compliance. In the body, the 40% is measured against a traditional hub design, not against SIMP or level-set. **Location:** Abstract (first quantitative comparison with a classical method, §5.1 rule 1). The number reappears in the closing summary paragraph of 'Method validation' (the article has no separate Conclusion section) and in the wheel-hub case study (Fig. 13, weight bar chart)

1. **R1: fail** (R1-unmatched (no classical design result; quality stratum)). Abstract vs body: the claimed advantage over SIMP and level-set has no classical counterpart in the article. No SIMP or level-set design, volume or compliance is reported (full text read; Table 6 compares only with ML references [36]-[38]). The 40% figure compares the optimized hub with a traditional hub of unstated origin (Fig. 13), not with a classical optimizer at matched volume or compliance, and the compliance of neither hub is quantified. The article's only quantitative SIMP comparison is a time (Fig. 12, 2.1 s vs 18.4 s) made with no quality comparison. Alternative: na, if R1 is read as applying to speed comparisons only; the §11 stratum-Q clarification (2026-09-24) makes R1 apply as written, so na is not used.
2. **R2: na** (R2-na-quality-stratum). Quality stratum, and no classical design is run, so the competence of a comparator cannot be judged. Under the §11 stratum-Q rule only criterion (b) could apply: the abstract names SIMP and level-set, while the body reports no stronger baseline for weight (the 40% is against a traditional hub, Fig. 13), so (b), a weaker baseline in the abstract than in the results, is not observed. For the Fig. 12 speed alternative: the SIMP time (about 18.4 s) is attributed to ref. [36], with no implementation, solver, iteration count, problem size or hardware for either method, so (a)-(e) cannot be established; partial (R2-hardware-unjustified) would be the likely code under that reading [inferred].
3. **R3: na** (R3-na-quality-stratum). Primary outcome is quality only. For C3: the PPO agent's training (100 episodes of FEA-in-the-loop interaction, Table 4, 'Method validation') has no reported time or hardware, and the 2.1 s inference time in Fig. 12 excludes it; no break-even is given.
4. **R4: partial** (R4-no-failure-rate). Step 4 and Table 5: the final ToD design is said to be checked by FEniCS FEA; volume fraction 0.32 and constraint values (0.49 mm, 298 MPa) are reported. The returned binary grid has disconnected and floating elements ('Method validation', Fig. 7), which the authors say SDF smoothing removes, but they are not counted, and Table 5 does not say whether it describes the design before or after smoothing. One design per case, so no failure rate. The 40% hub figure has no FE-computed compliance for either hub.

P01971 · Tskhondia (2025)

Reinforcement Learning Guided Engineering Design: from Topology Optimization to Advanced Modelling. Journal of Next-Generation Research 5.0 (JNGR 5.0), vol. 1, issue 3 (March-April 2025), ISSN 3075-2868; published 2025-03-12 (journal landing page); CC BY-SA per OpenAlex. DOI 10.70792/jngr5.0.v1i3.95. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On a 10×10 cantilever grid, refining a genetic-algorithm layout (about 1 min) with PPO (about 32 min) lowers the compliance metric from 19.3654 to 18.2786; the author states

GA alone does worse for the same wall-clock time. **Location:** Sec. 3, pp. 9-10 (first quantitative comparison with a non-learned method; the abstract states the GA comparison only qualitatively, the introduction has none, and there is no conclusion section: the closing Discussion restates the GA claim qualitatively)

1. **R1: partial** (R1-loose). Sec. 3, pp. 9-10: the printed numbers compare the GA output after about 1 min with GA+RL after about 1 + 32 min, so the budgets are unequal. The equal-wall-clock GA-alone result is asserted but not printed. GA generations, population and stopping rule are not stated, and the material volume of neither design is reported.

2. **R2: pass** (R2-none-established). Sec. 3: the GA and the RL refinement run on the same Apple M1 Pro with the same FE code [23]; the grid has ≈242 DOFs [derived], so (c) does not apply; the abstract gives no baseline figure (b); the GA ran shorter, not longer, than needed (d); no per-iteration time exists (e). A GA is a weak TO method compared with gradient TO (the appendix SIMP run takes about 1 s), but no R2 marker covers the choice of method. Unsure, so pass.

3. **R3: na** (R3-na-per-problem). No speed claim against a classical method is primary, and the RL agents are trained per problem (the PPO refinement of the 10×10 case is itself the 32 min run), which is R3's n/a case. Stratum Q (PROTOCOL §11, 2026-09-24) also makes R3 na. For C3: PPO 1M steps about 80 min on an i5 (Sec. 3); HRL hyperparameter search time not stated.

4. **R4: partial** (R4-no-constraint-report). Sec. 3: both designs are scored by the FE code of [23] in the same environment [inferred], one run each. Material volume, failed runs and seeds are not reported, and the metric is not defined. Removal-only refinement (Sec. 3, step 2) lowering compliance at a fixed load is inconsistent with pure compliance [inferred], so the metric may include a volume term. Fig. 3(c) (HRL, not the primary outcome) is labelled cherry-picked. A stricter reading gives fail.

P01974 · Ali (2026)

Toward Large Language Model-Driven Symbolic Topology Optimisation for Rapid Structural Concept Generation in Manufacturable Design. Journal of Manufacturing and Materials Processing 10(4), 117 (MDPI; published 30 March 2026; CC BY). DOI 10.3390/jmmp10040117. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). LLM-DSTO (local 8B LLM plus rule-based refinement) reaches about 87.5% of SIMP stiffness on a small 2D cantilever and 94.3% of SIMP thermal performance on a 30×30 heat sink, at claimed much lower cost without repeated FE simulations. **Location:** Abstract (quantitative quality ratios plus the qualitative cost claim); details in Sec. 4.1 and Fig. 7 (cantilever) and Sec. 4.2 and Fig. 10 (heat sink); repeated in Sec. 6

1. **R1: fail** (R1-unmatched). Abstract (p. 1), Secs. 4.1-4.2 (pp. 19-22), Figs. 7 and 10: the cost claim (no repeated FE simulations; about 1.5 s vs 5 s per iteration) is set against the SIMP reference designs (stopping rule not stated; taken as the finished optimum [inferred]), while the LLM designs are measurably worse (stiffness 12.5% lower; heat sink described as 5.7% worse). No equal-quality or equal-time matching is attempted, no SIMP run time is reported, and the volume fractions of the compared designs are not given (the LLM design meets the target only through a soft penalty, Eq. 13).

2. **R2: partial** (R2-inefficient-impl [inferred]). Secs. 4.1-4.2 (pp. 19-22): SIMP implementation, solver, hardware, iteration count and stopping rule are not stated. The only classical cost figure is a generic 'around 5 s per iteration' for traditional TO (Sec. 4.2, p. 22), placed in the heat-sink section (30×30 grid, Fig. 8) without a stated size, implementation or hardware. Our strong baseline's median per-iteration time on a larger 64×64 cantilever is 4.5 ms with PARDISO (retest/logs/g2_benchmark_2026-09-24.log, line 51), so the quoted figure is about 1,000 times slower [inferred across size and physics]. Because the 5 s is not a timed run of the article's SIMP at a stated size and hardware, R2(e) is not applied as fail. (b), (c) and (d) are not established: the abstract names the same SIMP as the results, problems are far below 10^5 DOFs, and no iteration count is given.

3. **R3: na** (R3-na-per-problem). Sec. 4: a pre-trained general-purpose LLM (DeepSeek R1 0528 Qwen3 8B) is used zero-shot for each problem, with no task-specific training and no TO data generation, so learning cost sits inside R1/R2. The LLM's pre-training cost is not attributable to this article.

4. **R4: partial** (R4-no-constraint-report). Secs. 4.1-4.2 (pp. 19-22), Figs. 7 and 10: one returned design per case is compared by normalised stiffness or thermal resistance, but the evaluation method is not

stated (an FE analysis of the returned grid is presumably needed for a stiffness on the SIMP scale [inferred]). The loop itself is driven by the heuristic score Q(X) (Secs. 3.3 and 3.9), but nothing states that the reported ratios are Q(X) values, and Fig. 6's symbolic-objective plateau (about 0.91) differs from the 0.875 stiffness ratio, so surrogate-only evaluation is not established. Achieved volume fractions are not reported (soft penalty only, Eq. 13), no failure rate or number of LLM runs is given, and the Fig. 10 axis label conflicts with the text's reading of the heat-sink result.

P01994 · Tskhondia (2025)

Practical topology optimization with deep reinforcement learning. ResearchGate item with a DataCite DOI; DataCite resourceType 'Preprint', publication year 2025; the author dates it December 2025 (ref. [5] of pool record P01997). OpenAlex types it 'article' with no venue. DOI 10.13140/rg.2.2.16548.95369. Stratum: none; access: no access; stage: adjudicated.

Primary outcome (paraphrase). Not coded (no full text and no abstract available). Only the title is known: deep reinforcement learning applied to topology optimization with a practical focus. **Location:** title only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P02042 · Feng et al. (2026)

DQN-Based Pre-Optimization for Dual-Scale Collaborative Topology Optimization of Anisotropic Materials. Applied Sciences 16(9), 4080 (MDPI; published 22 April 2026; open access, CC BY). DOI 10.3390/app16094080. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On the MBB beam, running the dual-scale thermo-mechanical optimization from a DQN pre-optimized start gave final compliance 0.5754 versus 0.7421 for the same framework without DQN pre-optimization, 22.463% lower. **Location:** Abstract (quantitative comparison with the traditional method); values in Sec. 5.1 (p. 17) and the Introduction (p. 2)

1. **R1: partial** (R1-loose). Sec. 5.1 (pp. 16-17), Table 4, Figs. 7-8: both designs share the mesh, loads and nominal volume fraction 0.4. Matching is loose: the baseline's penalty value, filter, iteration count and stopping rule are not stated; it ends with many intermediate densities (Fig. 7a), so a penalized grey field is compared with a near-binary design [inferred]; achieved volumes are not reported; which of the eight DQN runs (Table 3) gives 0.5754 is not stated; no compute budget is compared.
2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) reports the 22.463% compliance reduction against 'traditional methods', which is the same without-DQN run that Sec. 5.1 (pp. 16-17) and Fig. 8 report; the article has no appendix and no other classical baseline, so (b) is not observed. No time, solver or hardware is reported for the baseline. The baseline's unstated penalty value and grey final layout (Fig. 7a, p. 17) bear on R1, not on R2 in stratum Q.
3. **R3: na** (R3-na-quality-stratum). Quality stratum with per-problem online DQN training and no offline dataset (Sec. 3.4); no speed claim is quantified. For C3: no DQN training time, pre-optimization time or TO time is reported.
4. **R4: partial** (R4-no-constraint-report). Sec. 5.1-5.5: the compliance values are the optimizer's FE results for the returned designs [inferred]. Achieved volume fractions are not reported for any design, and failed or non-converged runs are not counted. Eight seeds are reported for the DQN method (Table 3), while the baseline is a single run. The 3D extrusions (Fig. 12) are not analysed (Sec. 5.6).

P02074 · Bhagat et al. (2026)

UDHF-SMERO: a unified deep hybrid framework for co-evolutionary smart material design and energy-adaptive structural optimizations. Journal of Building Pathology and Rehabilitation 11:165 (Springer; received 14 Nov 2025, published online 23 May 2026; CC BY-NC-ND 4.0). DOI 10.1007/s41024-026-00833-7. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). The framework's topology-to-performance mapping reaches an average R^2 of 0.96 over three load scenarios, versus 0.82, 0.84 and 0.86 for Method [3] (a cited BESO bridge study), Method [8] and Method [25] (Table 2). **Location:** Sec. 5 Conclusion (first quantitative comparison: superiority with average R^2 0.96, citing Table 2); Table 2, p. 11. The abstract reports only the method's own metrics; the introduction has no comparison of this method

1. **R1: partial** (R1-loose). Sec. 4, p. 11 and Table 2: the comparison is said to use the same conditions and data split (70/15/15). No implementation, budget, stopping rule or hardware is given for Methods [3], [8] or [25], which the reference list gives as a BESO pedestrian-bridge study, an evolutionary-optimization smart-grid study and a conformal-antenna study (refs. 3, 8, 25, pp. 19-20). A BESO run [3] has no performance-prediction R^2 [inferred], and the article does not say how any comparator was applied to its dataset, so the matching cannot be checked. This is the §5.3 partial case (quality reported, matching loose); no R1 fail criterion concerns a quality-only claim.
2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (p. 1) reports only the framework's own metrics and names no baseline, so (b) cannot apply. For the record: the comparators are cited, not described (no implementation, solver or hardware, Sec. 4, p. 11), and Sec. 4.4 (p. 16) gives 0.34 s per sample for unnamed traditional TO models with no method, size or hardware.
3. **R3: na** (R3-na-no-speed-claim). The primary outcome is prediction accuracy, and Sec. 4.4 concedes a slower pipeline (0.82 s vs 0.34 s per sample). For C3: no training time or data-generation cost is reported; only epoch counts and a 35% training-time reduction from mixed precision (Sec. 4, 4.4).
4. **R4: fail** (R4-similarity-metric). Table 2 and Sec. 5: quality rests on agreement scores, i.e. R^2 between predicted and FEM performance and reconstruction accuracy (%) in Tables 2 and 4. No FE-computed objective, volume or constraint satisfaction of returned designs is reported, and no failed designs are counted. The case study (Sec. 4.7) gives only agreement percentages.

P02084 · Ignuta-Ciuncanu et al. (2026)

Hierarchical thermal design for battery packs using phase change materials. International Journal of Heat and Mass Transfer (2026), article 129081; hybrid open access, CC BY-NC-ND per OpenAlex. DOI 10.1016/j.ijheatmasstransfer.2026.129081. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no full text). The abstract says a generative latent-space fin design needs over 95% fewer design variables than pixel-wise TO, and claims gains in thermal performance and computational efficiency; no time comparison is given. **Location:** abstract only

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P02128 · Jadhav et al. (2024)

Generative Lattice Units with 3D Diffusion for Inverse Design: GLU3D. Advanced Functional Materials 34(41) (2024) 2404165 (Wiley; published online 2024-06-06, issue dated October 2024, per Crossref); hybrid open access, CC BY-NC 4.0, per Crossref, OpenAlex and Semantic Scholar. DOI 10.1002/adfm.202404165. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (no accessible full text). Abstract: a 3D diffusion model generates lattice unit cells for target stiffness, said to speed up design and to beat simple-cubic and BCC cells in energy absorption and compression load at lower density. **Location:** Abstract (the only text read)

1. **R1: na** (no marker). no full text
2. **R2: na** (no marker). no full text
3. **R3: na** (no marker). no full text
4. **R4: na** (no marker). no full text

P02187 · Erzmann and Dittmer (2024)

Equivariant neural operators for gradient-consistent topology optimization. Journal of Computational Design and Engineering 11(3) (2024) 91-100 (Oxford University Press; advance access 19 April 2024; open access, CC BY 4.0). DOI 10.1093/jcde/qwae039. Stratum: speed; access: full text; stage: adjudicated.

Primary outcome (paraphrase). On SELTO disc test problems (39×39×4 voxels), SIMP with a learned neural-operator PDE solver runs in 2-19 s versus 75 s with the classical finite-difference solver on the same Xeon Silver 4210 CPU, stated as at least fourfold faster. **Location:** Sec. 5.3, p. 96, and Table 2, p. 97 (main text; the abstract, introduction and conclusion hold no quantitative classical comparison). Chosen under §5.1 rule 4 over the earlier Table 1 (IoU against the SIMP designs, p. 96), because the article states its focus is a speed-up of classical SIMP (p. 96)

1. **R1: partial** (R1-loose). Tables 1-2 and Sec. 5.3: both runs use the same 60-iteration SIMP loop [inferred], and quality is reported, but the learned-solver designs match the SIMP designs only in part (best IoU 0.73) and 6.25-73.75% of them fail the stress or connectivity check, while the SIMP reference's fail% and compliance are not given. No matching is attempted; the authors acknowledge a quality-speed trade-off (p. 96). If these numbers are read as measurably worse than SIMP, R1-unmatched (fail) applies.
2. **R2: partial** (R2-inefficient-impl). Table 2 (p. 97) and Sec. 5.3 (p. 96): the FDM SIMP takes 75 s for 60 iterations at about 2.4×10^4 DOFs, about 1.25 s per iteration [derived], one FDM solve at 160,000 voxels takes almost 5 min, and the authors state that FDM cost grows quadratically with the voxel count. The article does not name the linear solver. DL4TO's public source (dl4to/pde.py, read 2026-09-24; outside the article) solves the FDM system with SciPy's general sparse LU (UMFPACK if installed, else SuperLU), not a Cholesky/LDL^T or multigrid solver; in our G2 log a SciPy-SuperLU SIMP needs 48.6 ms per iteration against 4.5 ms for PARDISO at 64×64 (retest/logs/g2_benchmark_2026-09-24.log, lines 51 and 77). The implementation thus looks slow for the size [inferred], but no 3D re-test exists, so no direct comparison is possible. Both methods run on the same CPU, so (c) does not apply; (b) not observed; (d) not established (no convergence criterion or plot). Pointer for (e): 1.25 s per iteration at 39×39×4 [derived], to be compared with our strong baseline at the same size.
3. **R3: partial** (R3-no-data-cost). Sec. 5.1 and 5.3: training time is reported (2-14 h) and the data need is discussed only qualitatively; the data-generation cost is not reported and no break-even is given. For C3 [derived]: 200 trajectories at the 75 s average (15,000 s) plus 7,200-50,400 s of training, against a saving of 56 s per run for the equivariant U-Net FNO (75 − 19), break even after about 400-1,170 runs; for the fastest model (73 s saved), after about 300-900 runs.
4. **R4: pass** (R4-FE-verified). Sec. 5.2 and Table 1: the final design of every test problem is scored by IoU, compliance and a failure check (stress above 1.1× yield, or loaded voxels disconnected from the supports), and failures are counted (fail%). The article does not say which solver scores the designs [classical solver inferred]; volume fractions are not reported. Unsure, so pass.

P02296 · Liu et al. (2026)

Co-Optimization of Structure and Manufacturable Semi-Continuous Layers for Laminated Composites. ACM Transactions on Graphics 45(4), article 146 (SIGGRAPH 2026; published 3 July 2026 per the Manchester repository; CC BY per OpenAlex). DOI 10.1145/3811393. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract reports that jointly optimized fabric-reinforced laminates are up to 43.8% stiffer than counterparts made with planar fabric plies, in simulation and physical tests; no classical-optimizer comparison is stated. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no-full-text). no full text
2. **R2: na** (no-full-text). no full text
3. **R3: na** (no-full-text). no full text
4. **R4: na** (no-full-text). no full text

P02361 · Krischer et al. (2024)

Distributed design optimization of multi-component systems using meta models and topology optimization. Structural and Multidisciplinary Optimization 67:160 (published online 12 Sep 2024; CC BY 4.0). DOI 10.1007/s00158-024-03836-5. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). Designs from the meta-model-based decoupled architecture (system optimization with learned feasibility and mass estimators, then independent component TO) are feasible but up to 12.9% heavier than monolithic TO of the whole system. **Location:** Abstract (quantitative comparison with a classical method); numbers in Table 5, p. 16 (P2, 12.9%) and Table 3, pp. 12-13 (P1, 3.87-8.18%); restated in Sec. 8

1. **R1: partial** (R1-loose). Tables 3 (p. 12) and 5 (p. 16): both approaches solve the same problems with the same design domains, P1 mesh, material and 1 mm displacement limit, and both designs meet the limit (0.99-1.00 mm), so mass is compared at matched constraints. Matching is still loose: the optimizer of the monolithic run (A) is not named (MMA is named only for the component problems, Sec. 3.2, p. 7), no stopping rule or iteration count is stated for (A) or the component runs, the P2 mesh is not stated (Sec. 6.1), and the stochastic PSO system optimization of (B) is run once per case with no seeds. An unstated classical stopping rule is the §5.3 R1 partial example. The comparison disfavors the ML approach.
2. **R2: na** (R2-na-quality-only). No runtime of either approach is reported (Sec. 7), and no speed claim is quantified, so the efficiency criteria (a)-(e) do not apply. The monolithic baseline is the authors' own MATLAB SIMP formulation on the same model.
3. **R3: na** (R3-na-no-speed-claim). No quantified speed claim. For C3: neither sampling (component TO runs) nor meta-model training cost is reported; Sec. 7 calls the offline database computationally expensive and says training is excluded from the time argument.
4. **R4: pass** (R4-FE-verified). Tables 3 and 5: mass and end-effector displacement are reported for the returned component designs of both approaches. The displacement is computed from the condensed FE stiffness of the detailed designs [inferred from Sec. 3.2 and Eqs. 6-8]. Constraint satisfaction is reported for every case (0.99-1.00 of the limit), and all final designs are feasible (Sec. 5.4, pp. 11 and 14; Sec. 6.4, p. 17). The classifier's error rates are reported (FPR, TPR, accuracy; Tables 2 and 4), and a failed temporary classifier during sampling is disclosed (Sec. 6.2).

P02476 · Wang et al. (2020)

Deep generative modeling for mechanistic-based learning and design of metamaterial systems. Computer Methods in Applied Mechanics and Engineering 372 (2020) 113377 (journal reference on the arXiv record; bronze open access per OpenAlex). Preprint: arXiv 2006.15274 (primary cs.CE; also cs.LG, stat.ML; v1 27 Jun 2020, v2 16 Sep 2020). DOI 10.1016/j.cma.2020.113377. Stratum: quality; access: full text; stage: adjudicated.

Primary outcome (paraphrase). For the smiley-face target-displacement problem, the authors report that single-scale SIMP tends to give unrealistic designs with many intermediate densities, whereas the VAE-based two-stage framework yields a connected aperiodic structure matching the target (RRMSE 0.1623). **Location:** Sec. 5.2, p. 34 (the only classical run mentioned; described in words, no SIMP result shown). The abstract makes no classical comparison; the efficiency statements on p. 34 and in the conclusion (p. 36) rest on no classical run, time or cited cost

1. **R1: partial** (R1-loose). p. 34: the SIMP run is described in one sentence; its formulation, volume, objective value and design are not reported, so the comparison cannot be matched or checked. No speed comparison is made.

2. **R2: na** (R2-na-quality-only). Stratum Q (PROTOCOL §11, 2026-09-24): R2 applies only through criterion (b). The abstract (pp. 1-2) names no classical baseline, so (b) cannot apply. For the record: no classical time, implementation or hardware is reported anywhere (Sec. 5.2, pp. 31-34).

3. **R3: na** (R3-na-quality-stratum). Stratum Q (PROTOCOL §11, 2026-09-24): R3 is na. The efficiency statements (p. 34; Sec. 6, p. 36) are not measured against a classical method. For C3: VAE training took 44,496 s (p. 14); the cost of generating and homogenizing about 250,000 microstructures is not reported (pp. 13-14).

4. **R4: partial** (R4-no-failure-rate). Figs. 15-17: the assembled full structures (300,000 and 6.25 million fine-scale elements) are loaded and their displacement profiles compared with the targets by RRMSE [FE analysis of the returned structure inferred from the element counts]; boundary compatibility is shown visually. Decoder outputs are thresholded at 0.9 and cleaned of isolated pixels (p. 16). No failure or incompatibility counts are reported, and the SIMP design is neither shown nor evaluated.

P02479 · Garland et al. (2021)

Pragmatic generative optimization of novel structural lattice metamaterials with machine learning. Materials & Design 203 (2021) 109632 (Elsevier; publication date 2021-03-02 per OpenAlex; gold open access, CC BY per OpenAlex and Semantic Scholar metadata; OSTI record 1768685). DOI 10.1016/j.matdes.2021.109632. Stratum: none; access: abstract only; stage: agreed.

Primary outcome (paraphrase). Not coded (abstract only). The abstract says an ML-guided search finds Pareto-optimal lattice unit cells for stiffness under static load and low wave speed under impact, with about 3500 simulation calls. It names no classical comparator. **Location:** Abstract (pool record, from OpenAlex)

1. **R1: na** (no marker). no full text

2. **R2: na** (no marker). no full text

3. **R3: na** (no marker). no full text

4. **R4: na** (no marker). no full text